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Debapt: Ontology-driven multi-agent debate for APT adversary profile construction from cyber threat intelligence

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ABSTRACT

Cyber threat intelligence (CTI) reports contain rich information about advanced persistent threat (APT) groups. This information is useful for adversary profile construction. However, turning long and fragmented CTI reports into structured adversary profiles remains difficult. Existing methods mainly rely on named entity and relation extraction pipelines or single large language models (LLMs). These methods often lack explicit semantic constraints for profile-oriented tasks. They are also prone to omission and hallucination when processing long reports. In this paper, we study APT adversary profile construction from CTI reports. We formulate this task as an ontology-constrained profile information extraction task. To support this task, we develop VICTOR, a domain-specific ontology that organizes profile-relevant information into six dimensions. We then propose Debapt, an ontology-driven multi-agent debate framework. Guided by VICTOR, Debapt performs APT adversary profile construction through two debate loops in the entity extraction phase and relation extraction phase. In each loop, role-specialized agents iteratively extract, review, and adjudicate candidate profile-relevant facts under moderator supervision. These extracted results can be further aggregated across reports to support actor-centric adversary profile construction. To evaluate Debapt, we re-annotate three public CTI datasets under a unified profiling schema. Experimental results show that Debapt improves profile-oriented entity and relation extraction over competitive baselines. It extracts more complete and better grounded profile-relevant facts from CTI reports. Case studies further show that these extracted results can support the construction of analytically useful adversary profiles.

1. Introduction

As advanced persistent threat (APT) campaigns increasingly span multiple stages, evolve over time, and persist for extended periods, cyber defense can no longer rely solely on isolated indicators or signature matching (Tijjani et al., 2026; Salim et al., 2023; Shen et al., 2022). Defenders instead need to connect information on infrastructure, malware payloads, and adversary tactics, techniques, and procedures (TTPs) to reconstruct actor-level behavior (Irshad and Siddiqui, 2023). In this context, APT adversary profile construction provides an actor-centric way to organize dispersed information into a coherent analytical view (Prasad et al., 2025). By systematically integrating information such as infrastructure usage, malicious capabilities, and TTP preferences, it helps analysts better understand adversary behavior and supports long-term tracking and comparison (Bjurling and Raza, 2024; Ahmed et al., 2022).

CTI reports are a primary source for APT adversary profile construction (Rani et al., 2024; Ma et al., 2025). They contain technical observations, incident analysis, and historical context collected from real-world investigations (Wagner et al., 2019). In practice, many key profile elements, such as victim targets, infrastructure, malware, and TTPs, are recorded in CTI reports before they appear in structured databases (Sun et al., 2023; Stănciulescu et al., 2024). However, CTI reports are typically lengthy, context-rich, and written in unstructured natural language, with critical information dispersed across paragraphs, sections, and narrative details (Huang et al., 2024; Husari et al., 2017; Chen et al., 2024). This makes it difficult to directly transform report

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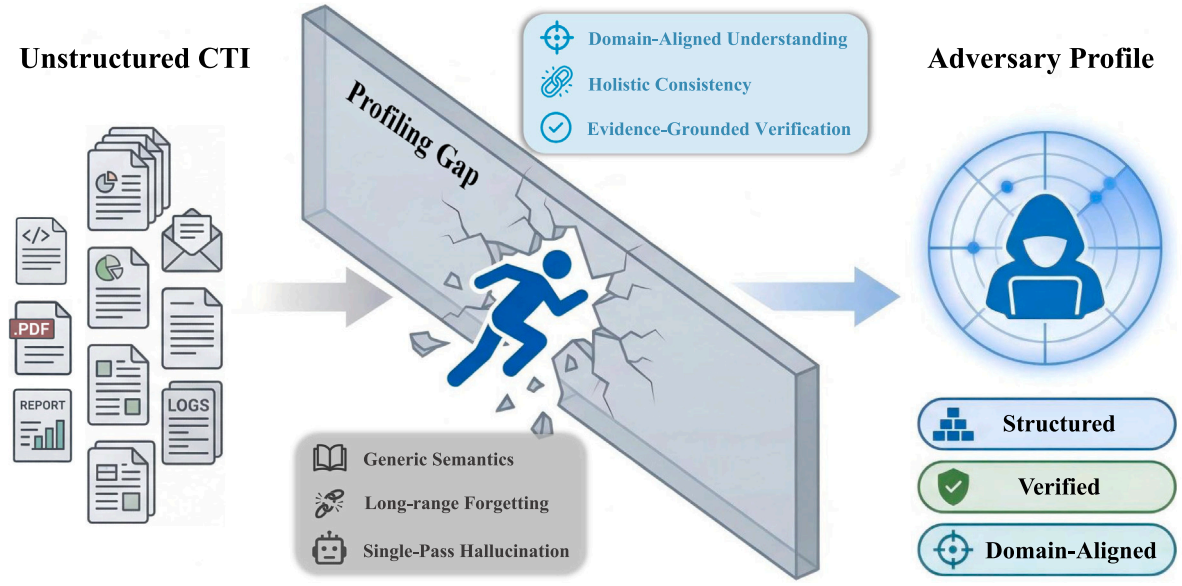


Fig. 1. Conceptual illustration of the profiling gap in APT adversary profile construction. (Left) Raw CTI reports contain rich but fragmented and unstructured information. (Center) Existing extraction strategies are limited by generic semantic understanding, long-range forgetting, and single-pass hallucination. Bridging this gap requires a profile-oriented strategy with domain-aligned understanding, holistic consistency, and evidence-grounded verification. (Right) This shift supports the transformation of scattered CTI information into adversary profiles that are structured, verified, and domain-aligned.

content into structured adversary profiles, limiting the timeliness and systematicity of security analysis (Li et al., 2022).

Existing research mainly follows two directions for APT adversary profile construction from CTI reports. The first uses named entity recognition and relation extraction pipelines, often with rule-based methods or deep learning models, to extract threat information from CTI reports (Morwal et al., 2012; Bridges et al., 2017; Aravind et al., 2023). The second uses a single large language model (LLM) to extract information through zero-shot or few-shot prompting (Cheng et al., 2025; Zhang et al., 2025). Although these methods improve automation, they still face three practical challenges, as illustrated in Fig. 1. First, CTI reports contain specialized terminology, implicit references, and domain-specific semantics (Aghaei et al., 2025). Extraction methods based on simple rules or local semantic cues often struggle to map dispersed descriptions to profile needs in a precise and consistent way (Jo et al., 2022; Peng et al., 2026). Without explicit profile-oriented semantic constraints, this can weaken domain alignment and lead to unstable profile semantics **(Challenge 1)**. Second, CTI reports are often long and context-rich. Important information may appear in distant paragraphs or different sections. Traditional extraction can lose these long-range links (Hilgert et al., 2024), which leads to missing facts and broken cross-paragraph associations **(Challenge 2)**. Third, generative models are prone to hallucination when handling precise cyber facts such as IP addresses, hash values, file paths, and malware artifacts (Xu et al., 2024; Froudakis et al., 2025). Without explicit verification, such errors can directly reduce the reliability of the constructed profile **(Challenge 3)**.

To address these challenges, we propose an ontology-driven framework that leverages multi-agent **deba**te to automate **APT** adversary profile construction without task-specific fine-tuning, named **Debapt**. Our key idea is to formulate APT adversary profile construction as an ontology-constrained profile information extraction task over CTI reports. We first develop VICTOR, a profile-oriented ontology that organizes profile-relevant adversary information into six dimensions and provides explicit semantic constraints for APT adversary profile construction. Based on VICTOR, Debapt enables agents to focus on specific dimensions within the VICTOR view through role division. This design mitigates information forgetting in long CTI report processing by reducing the task load in each extraction step. On this basis, Debapt performs APT adversary profile construction through two debate loops in the entity extraction phase and relation extraction phase. Each phase adopts an extract-review-adjudicate debate process in which agents iteratively cross-check intermediate outputs against source text and schema constraints. This design helps reduce information loss in long documents, improves grounding to the original text, and mitigates hallucination during profile construction. The main contributions are as follows:

- We develop VICTOR, a profile-oriented ontology that defines six core dimensions and provides a unified semantic space for APT adversary profile construction from CTI reports.
- We propose Debapt, an ontology-driven multi-agent debate framework for APT adversary profile construction through the entity extraction phase and relation extraction phase.
- We re-annotate open-source datasets for APT adversary profile construction and conduct extensive experiments showing the effectiveness of Debapt. A case study further demonstrates the practical value of the framework in real-world security analysis.

In this paper, we study APT adversary profile construction, which can also be referred to as adversary profiling when no ambiguity arises. The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 defines the problem setting. Section 4 elaborates on the VICTOR ontology and the Debapt framework. Section 5 presents the experimental setup, quantitative results, and case study findings. Section 6 concludes the paper and discusses future research directions.

2. Related work

In this section, we review related work in three branches: (1) Ontology; (2) CTI knowledge extraction; (3) Multi-agent debate framework.

2.1. Ontology

An ontology is a formalized conceptual model that describes domain knowledge. It abstracts core concepts within a domain into classes, characterizes their essential features through properties, and defines

semantic associations among different concepts using relations. Based on this schema, specific objects within the domain are represented as instances of these classes. These objects are systematically recorded using property values and relationship links, generating a reusable and extensible knowledge representation (Meenachi and Baba, 2012).

The Zachman framework (Zachman, 2006) provides a two-dimensional method for knowledge organization. It systematically categorizes complex information using the 5W1H model. This structure helps reduce descriptive omissions and redundancy. Based on this, researchers propose various semantic models and frameworks for representing threat intelligence. General ontologies, including the STIX (STIX, 2023) model and UCO (Syed et al., 2016), cover a broad scope. They apply to general threat intelligence and unify diverse security data. Specific subdomains and tasks require dedicated ontologies. For example, MALOnt (Rastogi et al., 2020) acts as a specialized ontology for malware threat intelligence. It integrates knowledge from unstructured and heterogeneous sources into a single malware semantic framework. The MITRE ATT&CK framework (Mitre, 2026) and its extensions (Huang et al., 2025; Akbar et al., 2023) focus on modeling the tactical and technical semantics of adversary behaviors. These models support the alignment, querying, and inference of TTPs. Furthermore, Diamond (Caltagirone et al., 2013) proposes a conceptual model using events as atomic units from an intrusion analysis perspective, characterizing events through four elements.

Overall, while existing ontologies facilitate the structured representation of threat intelligence, a specialized ontology for APT adversary profile construction remains absent. Current models lack the capacity to organize profile-relevant infrastructure, capabilities, TTPs, and temporal information elements. This makes it difficult to directly support the construction of actor-centric adversary profiles.

2.2. CTI knowledge extraction

When knowledge is embedded within large volumes of text, it becomes difficult to utilize effectively. To address this challenge, methods have emerged for extracting threat intelligence from unstructured CTI reports.

Traditional named entity recognition methods generally fall into two categories: rule-based matching and machine learning approaches. Rule-based techniques rely on hand-crafted rules designed by experts. These methods identify entities by combining dictionaries with syntactic or lexical patterns (Etzioni et al., 2005; Bridges et al., 2017). Machine learning approaches typically perform sequence labeling using algorithms such as hidden Markov models (Morwal et al., 2012), support vector machines (Mansouri et al., 2008), perceptrons (Li et al., 2020), and conditional random fields (CRF) (Jia et al., 2018). Among these, conditional random fields serve as a standard sequence modeling technique. For example, Joshi et al. (2013) deployed a conditional random field model from Stanford NER to enhance security tasks related to zero-day attacks.

Within deep learning approaches, BiLSTM (Wang and Liu, 2023) networks capture contextual representations through bidirectional sequence modeling. Integrating CRF into this architecture further improves label sequence consistency by applying global decoding constraints (Chen et al., 2019). The development of pre-trained language models introduces BERT (Xu et al., 2019) for generating robust contextualized representations, leading to combinations like BERT-CRF and BERT-BiLSTM-CRF (Dasgupta et al., 2020). CyTIE (Aravind et al., 2023) demonstrated in the experimental results that the contextual embeddings of BERT effectively complement the sequence labeling capabilities of conditional random fields, confirming the optimal performance of the BERT-BiLSTM-CRF configuration. For specialized domains, SecureBERT (Aghaei et al., 2022) aligns closely with threat intelligence terminology through pre-training on large-scale security corpora, yielding strong results in downstream tasks. EXTRACTOR (Satvat et al., 2021) generates attack behavior graphs without relying on domain-specific textual assumptions. Neural network architectures such as TA-GCN (Shang et al., 2024) and KnowCTI (Wang et al., 2024) further advance entity and relation modeling by integrating domain knowledge or leveraging perceptual embeddings.

With the advancement of large language models, researchers have begun exploring their semantic understanding capabilities for CTI extraction. Methods relying on prompting treat these models as general extractors and structural generators. Approaches including CTIKG (Huang and Xiao, 2024), CTINexus (Cheng et al., 2025), and AttacKG+ (Zhang et al., 2025) convert CTI reports directly into knowledge graphs. They achieve this transformation through procedural constraints, in-context examples, and structured formatting. Conversely, methods based on fine-tuning adapt general models into specialized extractors. Hu et al. (2024) and Lekssays et al. (2025) fine-tune models using LoRA to extract entity relationships and construct knowledge graphs.

2.3. Multi-agent debate framework

With the advancement of LLMs, deploying multiple LLMs as agents and solving problems through collaborative discussion has become a common strategy (Guan et al., 2025). A typical multi-agent debate process requires agents to generate independent responses initially. These agents then question and revise the outputs of one another across multiple interactive rounds. This mechanism improves both reasoning capabilities and factual reliability (Du et al., 2024; Liang et al., 2024). Researchers further emphasize perspective diversity. Assigning distinct roles and viewpoints makes the discussion resemble human group peer review. This approach applies to evaluation scenarios where language models act as judges, ultimately enhancing assessment consistency (Chan et al., 2023). Other studies systematically design debates at the collaboration framework level. These works explicitly model teammates' perspectives in collaborative tasks (Li et al., 2023). This cognitive modeling helps mitigate information loss in long contexts and reduces state hallucinations. Increasing the number of agents and debate rounds introduces significant computational costs. Subsequent research proposes strategies such as sparse communication, selective retention of key information, and stability detection. These methods reduce operational overhead and prevent ineffective iterations (Li et al., 2024; Hu et al., 2025; Zeng et al., 2025). Researchers apply these concepts to specific downstream tasks. For example, multi-perspective debates improve the quality of event relation extraction, and multi-phase collaboration among diverse roles facilitates emotional support conversations (Xu et al., 2025).

3. Preliminary

We formulate APT adversary profile construction as an ontology-constrained profile information extraction task. Given a collection of unstructured CTI reports, the objective is to extract profile-relevant entities and semantic relations from text, and to organize them into a structured representation of the adversary described in each report.

Ontology-constrained semantic space. Let $\mathcal{D} = \{d_1, d_2, \ldots, d_n\}$ denote a collection of CTI reports, where each d_i is an unstructured document. Let $\mathcal{O}$ denote the profile-oriented ontology, referred to as VICTOR in this work. We define $\mathcal{O} = (\mathcal{M}, \mathcal{C}, \mathcal{R}, \tau, \mu, \mathcal{H})$, where $\mathcal{M}$ is the set of profile dimensions, $\mathcal{C}$ is the set of entity types, $\mathcal{R}$ is the set of relation types, and $\tau : \mathcal{C} \to \mathcal{M}$ maps each entity type to its profile dimension. The mapping function $\mu : \mathcal{C} \to 2^{\mathcal{S}}$ aligns VICTOR entity types with representative concepts in existing CTI standards $\mathcal{S}$, including STIX domain objects, STIX cyber observable objects, and MITRE ATT&CK concepts. The relation hint set $\mathcal{H} \subseteq \mathcal{C} \times \mathcal{R} \times \mathcal{C}$ specifies reasonable source and target entity type pairings for each relation type. In this work, $\mathcal{M}$ corresponds to the six dimensions of VICTOR. Together, these components define the semantic scope of APT adversary profile

construction. At the relation level, $\mathcal{H}$ is derived from STIX-compatible relation semantics and APT profiling needs, and is used as lightweight semantic guidance for relation extraction and review. It serves as soft guidance rather than a strict constraint. The ontology defines the semantic boundary of the task, the alignment with representative CTI standards, and the type-level prior knowledge used during profile extraction.

Document-level profile. For each report $d_i \in \mathcal{D}$, the target output is a document-level profile $\mathcal{G}_i = (\mathcal{V}_i, \mathcal{E}_i)$, where $\mathcal{V}_i$ denotes the set of entity instances extracted from d_i, and $\mathcal{E}_i$ denotes the set of semantic relations among them. Each relation is represented as a triple over the extracted entities:

$$\mathcal{E}_i \subseteq \mathcal{V}_i \times \mathcal{R} \times \mathcal{V}_i.$$

For any entity instance $v \in \mathcal{V}_i$, let $\mathrm{type}(v) \in \mathcal{C}$ denote its ontology type. An entity is retained only if it is supported by evidence in the source text and its assigned type is consistent with the ontology. Similarly, a candidate relation triple (h, r, t) is retained only if it is supported by textual evidence and its relation type is compatible with the STIX relation schema. To assist relation extraction and review, we further provide type-level hints in the form of *entity type → relation type → entity type*. These hints serve as lightweight references for common semantic pairings, rather than as strict constraints. As a result, even when a specific type combination is not explicitly covered by the hints, the relation can still be retained if it is appropriate in context and supported by the source text.

For a single report d_i, the profile construction process can be written as

$$F(d_i, \mathcal{O}) = \mathcal{G}_i,$$

where F denotes the ontology-constrained mapping from an unstructured CTI report to its document-level profile. At the corpus level, applying F to each report in $\mathcal{D}$ yields a collection of document-level profiles, denoted by $\mathcal{G} = \{\mathcal{G}_i \mid d_i \in \mathcal{D}\}$.

Debate-based iterative refinement. To realize the mapping $F(d_i, \mathcal{O}) = \mathcal{G}_i$, the profile construction process is modeled as a controlled multi-agent debate with two phases: entity extraction phase and relation extraction phase, denoted by $s \in \{\mathrm{ent}, \mathrm{rel}\}$. At round k, the state of phase s is defined as

$$S_{i,s}^{(k)} = \left(\hat{\mathcal{X}}_{i,s}^{(k)}, \mathcal{B}_{i,s}^{(k)}, y_{i,s}^{(k)}\right),$$

where $\hat{\mathcal{X}}_{i,s}^{(k)}$ is the candidate result set proposed in the current round, $\mathcal{B}_{i,s}^{(k)}$ is the review feedback, and $y_{i,s}^{(k)} \in \{\mathtt{accept}, \mathtt{revise}\}$ is the adjudication decision. We use $\mathcal{X}_{i,s}^{(k)}$ to denote the retained result set after adjudication at round k. When $s = \mathrm{ent}$, both $\hat{\mathcal{X}}_{i,s}^{(k)}$ and $\mathcal{X}_{i,s}^{(k)}$ are entity sets. When $s = \mathrm{rel}$, they are relation sets.

Each round follows an extract–review–adjudicate transition:

$$\hat{\mathcal{X}}_{i,s}^{(k)} = \mathrm{Ext}_s\left(d_i, \mathcal{O}, \mathcal{X}_{i,s}^{(k-1)}, \mathcal{B}_{i,s}^{(k-1)}\right),$$

$$\mathcal{B}_{i,s}^{(k)} = \mathrm{Crit}_s\left(d_i, \hat{\mathcal{X}}_{i,s}^{(k)}, \mathcal{O}\right),$$

$$\left(\mathcal{X}_{i,s}^{(k)}, y_{i,s}^{(k)}\right) = \mathrm{Jdg}_s\left(d_i, \hat{\mathcal{X}}_{i,s}^{(k)}, \mathcal{B}_{i,s}^{(k)}, \mathcal{O}\right).$$

Here, $\mathrm{Ext}_s(\cdot)$ denotes the extractor agent, which generates new candidates based on the report, the ontology, the retained results from the previous round, and the corresponding feedback. $\mathrm{Crit}_s(\cdot)$ denotes the reviewer agent, which examines the candidates from the perspectives of evidential support, ontology consistency, and relation-level semantic plausibility. $\mathrm{Jdg}_s(\cdot)$ denotes the adjudicator agent, which integrates the current candidates and the review feedback to determine whether the results should be accepted or revised. A moderator agent controls round progression and phase termination throughout the process.

A phase terminates when $y_{i,s}^{(k)} = \mathtt{accept}$, when a predefined maximum number of rounds is reached, or when further iterations no longer yield valid corrections. The entity extraction phase outputs the final entity set $\mathcal{V}_i$. The relation extraction phase then outputs the final relation set $\mathcal{E}_i$ conditioned on $\mathcal{V}_i$. The final output for report d_i is the document-level profile $\mathcal{G}_i = (\mathcal{V}_i, \mathcal{E}_i)$.

4. Approach

4.1. Framework overview

Building on the problem formulation in Section 3, Debapt addresses APT adversary profile construction as an ontology-constrained profile information extraction task, rather than a one-pass extraction problem. The central challenge is that profile-oriented CTI understanding is inherently coupled: entity identification, semantic typing, relation extraction, and actor-centric organization are mutually dependent, while the supporting information is often dispersed across long and unstructured CTI reports. Under such conditions, direct single-model extraction is prone to omission, semantic drift, and hallucinated profile facts.

To address this challenge, Debapt integrates two complementary components. The first component is the VICTOR ontology, which specifies the categories of information to be extracted, clarifies their meanings and semantic boundaries, and provides a structure for organizing them. The second is a controlled multi-agent debate framework, which operationalizes these constraints through iterative extraction, review, and adjudication. In this design, the ontology defines the semantic space, whereas the debate framework progressively stabilizes candidate outputs through evidence grounding, review, and feedback-driven revision.

Accordingly, Debapt does not directly map a CTI report to a final profile in a single generation step. It decomposes the extraction process into two debate loops in the entity extraction phase and relation extraction phase. The first loop extracts profile-relevant entities under ontology guidance; the second extracts semantic relations over verified entities under lightweight relation constraints. Through this phase-based design, Debapt reduces long-context omission, limits error propagation across phases, and improves the reliability of the structured profile. Meanwhile, information from multiple reports can be further aggregated in the profile aggregation step to support actor-centric adversary profiles.

4.2. VICTOR ontology

A central requirement of adversary profiling is that the extracted information should be organized around the threat actor. Debapt is grounded in VICTOR, a domain-specific ontology tailored to adversary profiling. Unlike general-purpose security ontologies, VICTOR is designed to capture the semantic composition of an adversary profile, so that heterogeneous CTI information can be organized into a unified actor-centric representation. As formalized in Section 3, VICTOR is represented as $\mathcal{O} = (\mathcal{M}, \mathcal{C}, \mathcal{R}, \tau, \mu, \mathcal{H})$, which defines its profile dimensions, entity types, relation types, type-to-dimension mapping, standard alignment, and type-level relation hints.

Concretely, VICTOR organizes profile-relevant information into six dimensions (Fig. 2). The Victim dimension characterizes the targeted entities and their attributes, answering who is attacked. The Infrastructure dimension captures the network and system resources that support malicious campaigns, answering what resources are utilized. The Capability dimension outlines the technical tools and proficiencies employed by the attackers, answering what capabilities are used. The TTPs dimension captures the tactical and technical behaviors exhibited during the attack, answering how the attack is executed. The Originator dimension identifies the threat actors and their background attributes, answering who initiates the attack. The Record dimension tracks the temporal data concerning the threat activities, answering when the

Algorithm 1 Debapt: Ontology-Driven Multi-Agent Debate for Document-Level Profile Construction

Require: **CTI Reports** $\mathcal{D}$, **Profile-oriented Ontology** $\mathcal{O}$, **Maximum Debate Rounds** K
Ensure: **Document-Level Profile Set** $\mathcal{G}$

$\mathcal{G} \leftarrow \emptyset$
for each report $d_i \in \mathcal{D}$ **do**
  // **Phase A: entity extraction phase**
  $\mathcal{X}_{i,\text{ent}}^{(0)} \leftarrow \emptyset$, $\mathcal{B}_{i,\text{ent}}^{(0)} \leftarrow \emptyset$, $y_{i,\text{ent}}^{(0)} \leftarrow$ `revise`
  $k \leftarrow 1$
  while $k \le K$ **do**
    $\hat{\mathcal{X}}_{i,\text{ent}}^{(k)} \leftarrow \text{Ext}_{\text{ent}}\left(d_i, \mathcal{O}, \mathcal{X}_{i,\text{ent}}^{(k-1)}, \mathcal{B}_{i,\text{ent}}^{(k-1)}\right)$
    $\mathcal{B}_{i,\text{ent}}^{(k)} \leftarrow \text{Crit}_{\text{ent}}\left(d_i, \hat{\mathcal{X}}_{i,\text{ent}}^{(k)}, \mathcal{O}\right)$
    $(\mathcal{X}_{i,\text{ent}}^{(k)}, y_{i,\text{ent}}^{(k)}) \leftarrow \text{Jdg}_{\text{ent}}\left(d_i, \hat{\mathcal{X}}_{i,\text{ent}}^{(k)}, \mathcal{B}_{i,\text{ent}}^{(k)}, \mathcal{O}\right)$
    if $y_{i,\text{ent}}^{(k)} =$ `accept` **or** $\text{No_Valid_Correction}\left(\mathcal{X}_{i,\text{ent}}^{(k)}, \mathcal{X}_{i,\text{ent}}^{(k-1)}\right)$ **then**
      break
    end if
    $k \leftarrow k + 1$
  end while
  $\mathcal{V}_i \leftarrow \mathcal{X}_{i,\text{ent}}^{(k)}$
  // **Phase B: relation extraction phase**
  $\mathcal{X}_{i,\text{rel}}^{(0)} \leftarrow \emptyset$, $\mathcal{B}_{i,\text{rel}}^{(0)} \leftarrow \emptyset$, $y_{i,\text{rel}}^{(0)} \leftarrow$ `revise`
  $k \leftarrow 1$
  while $k \le K$ **do**
    $\hat{\mathcal{X}}_{i,\text{rel}}^{(k)} \leftarrow \text{Ext}_{\text{rel}}\left(d_i, \mathcal{V}_i, \mathcal{O}, \mathcal{X}_{i,\text{rel}}^{(k-1)}, \mathcal{B}_{i,\text{rel}}^{(k-1)}\right)$
    $\mathcal{B}_{i,\text{rel}}^{(k)} \leftarrow \text{Crit}_{\text{rel}}\left(d_i, \mathcal{V}_i, \hat{\mathcal{X}}_{i,\text{rel}}^{(k)}, \mathcal{O}\right)$
    $(\mathcal{X}_{i,\text{rel}}^{(k)}, y_{i,\text{rel}}^{(k)}) \leftarrow \text{Jdg}_{\text{rel}}\left(d_i, \mathcal{V}_i, \hat{\mathcal{X}}_{i,\text{rel}}^{(k)}, \mathcal{B}_{i,\text{rel}}^{(k)}, \mathcal{O}\right)$
    if $y_{i,\text{rel}}^{(k)} =$ `accept` **or** $\text{No_Valid_Correction}\left(\mathcal{X}_{i,\text{rel}}^{(k)}, \mathcal{X}_{i,\text{rel}}^{(k-1)}\right)$ **then**
      break
    end if
    $k \leftarrow k + 1$
  end while
  $\mathcal{E}_i \leftarrow \mathcal{X}_{i,\text{rel}}^{(k)}$
  $\mathcal{G}_i \leftarrow (\mathcal{V}_i, \mathcal{E}_i)$
  $\mathcal{G} \leftarrow \mathcal{G} \cup \{\mathcal{G}_i\}$
end for
return $\mathcal{G}$

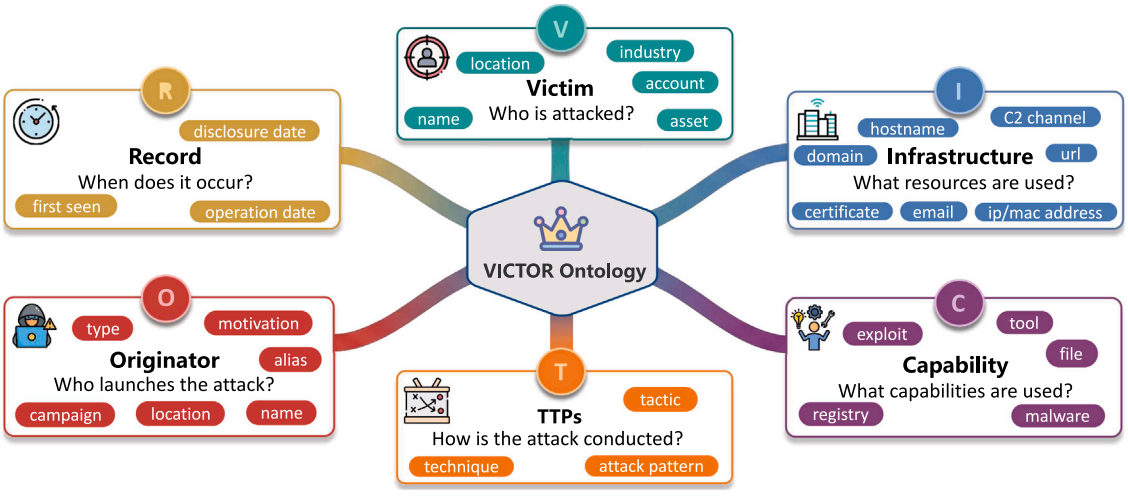


Fig. 2. Overview of the VICTOR ontology. VICTOR organizes profile-relevant CTI information into six dimensions, namely Victim, Infrastructure, Capability, TTPs, Originator, and Record, and provides an ontology-guided semantic space for APT adversary profile construction.

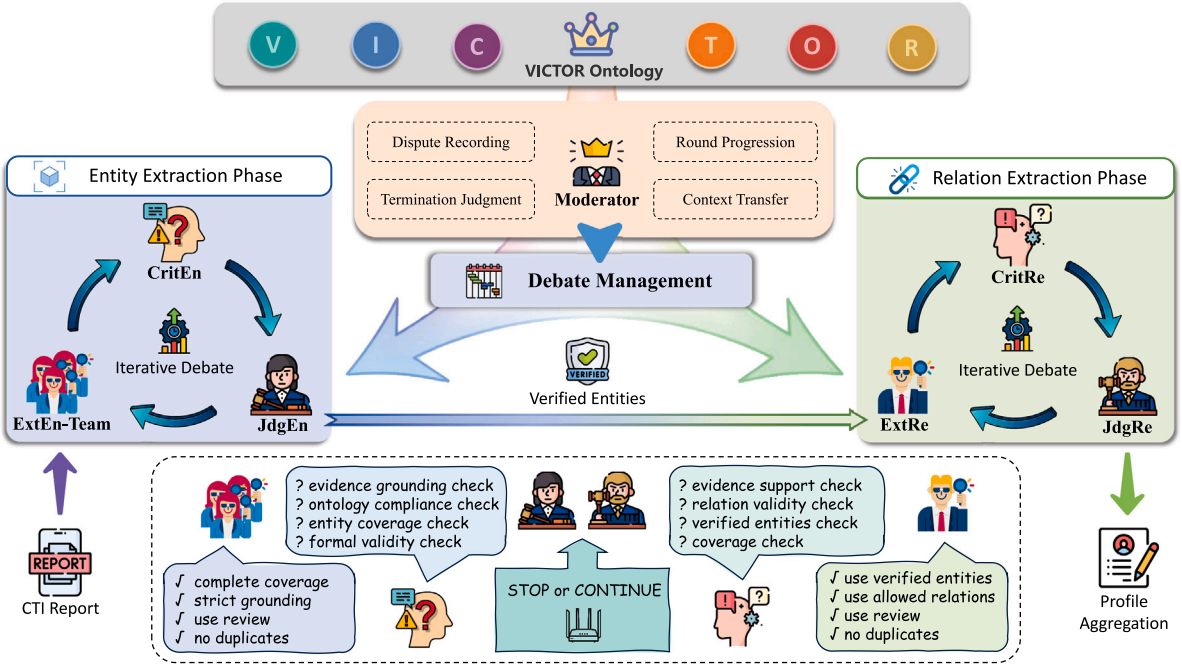


Fig. 3. Overview of the Debapt framework. Guided by the VICTOR ontology, Debapt performs adversary profiling through a two-phase multi-agent debate framework. The moderator coordinates the entity extraction phase and relation extraction phase to produce a document-level profile. A profile aggregation step can further aggregate information across reports to support actor-centric adversary profiles.

events occur and are publicly disclosed. For a complete explanation of the specific fields included in each dimension, see Appendix A Table 7. Appendix B further summarizes the alignment between VICTOR fields and representative STIX or MITRE ATT&CK concepts, showing both direct alignments and profiling-oriented semantic alignments. At the relation level, VICTOR adopts the STIX-compatible relation semantics, and provides type-level relation hints as a lightweight semantic constraint during relation extraction and review. Rather than strictly prescribing how entities must be connected, this schema provides a lightweight relation vocabulary together with soft guidance for identifying reasonable semantic links among extracted entities. The detailed type-level relation hints are provided in Appendix C.

This ontology serves three roles in the Debapt framework. First, it defines the semantic scope of adversary profiling, so that the extraction process remains focused on profile-relevant information. Second, it breaks complex CTI information into six profile dimensions, making long CTI report extraction more tractable and reducing confusion among unrelated types of profile-relevant information. Third, it provides a unified reference for review and adjudication, allowing candidate outputs to be assessed not only for textual support but also for semantic appropriateness under the ontology.

4.3. Multi-agent debate framework

Guided by the semantic constraints defined by VICTOR, Debapt performs adversary profiling through a controlled multi-agent debate framework (Fig. 3). This framework iteratively refines intermediate results until they become sufficiently grounded, semantically coherent, and stable enough for phase transition. In this sense, debate in Debapt serves as a feedback-driven correction mechanism for adversary profiling from CTI reports.

The framework is orchestrated by a moderator agent, denoted as Mod. Mod does not directly extract entities or relations. It acts as the global controller of the refinement process, managing round progression, passing intermediate context across roles, recording unresolved disputes, and deciding whether a phase should continue or terminate. This design separates procedural control from extraction behavior, which helps preserve consistency across rounds and keeps the multi-agent interaction focused and well controlled.

Operationally, Debapt comprises two staged debate loops. The first loop, entity extraction, aims to extract a stable ontology-aligned entity set from the source report. The second loop, relation extraction, operates over the verified entities and extracts semantic relations under relation schema guidance. Together, the two loops produce a document-level profile. The full pseudocode for the algorithm is provided in Algorithm 1. These report-level outputs can be further aggregated in a profile aggregation step to support actor-centric adversary profile construction across reports.

4.3.1. Entity extraction phase

The goal of the entity extraction phase is not merely to maximize the number of extracted mentions, but to establish a profile-relevant entity set that is evidence-grounded, ontology-consistent, and sufficiently comprehensive. This phase is carried out by three role-specialized agents: ExtEn, CritEn, and JdgEn.

ExtEn: ontology-guided extraction. Entity extraction from long CTI reports is vulnerable to omission because some of the clues are often distributed across distant paragraphs and expressed in diverse forms. To address this difficulty, ExtEn adopts a team-based design with six ontology-aligned prompts, each specialized to one VICTOR dimension. The prompt templates are shown in Appendix D Fig. 10 (Agent ExtEn Prompt). By decomposing the extraction task according to the ontology, ExtEn reduces interference across dimensions and improves the coverage of profile-relevant clues in long reports.

CritEn: evidence-based review and validation. High recall alone is insufficient for reliable adversary profiling. Candidate entities are therefore examined by CritEn through a review process from four perspectives. First, CritEn performs an evidence grounding check, examining whether each candidate entity is explicitly supported by the source CTI report. This step grounds the extraction in report evidence and suppresses hallucinated profile facts. Second, CritEn performs an ontology compliance check, verifying whether the assigned entity type is semantically compatible with the boundaries defined by VICTOR.

Third, CritEn conducts an entity coverage check by comparing the source report with the current candidate set, in order to identify profile-relevant elements that may have been omitted in earlier extraction. Fourth, CritEn incorporates formal validation for structurally constrained entity types. For example, IP addresses, MAC addresses, and CVE identifiers are examined using regular expressions, while tactic and technique entities are validated with a retrieval-augmented generation (RAG) (Lewis et al., 2020) mechanism to verify both identifier legality and name–ID consistency. Through this design, entity review is grounded in deterministic and retrieval-supported checks. The corresponding prompt templates are shown in Appendix D Fig. 10 (Agent CritEn Prompt).

JdgEn: adjudication and iterative refinement. The role of JdgEn is to determine whether the current entity state is stable enough to advance to the relation extraction phase. JdgEn synthesizes the extraction results and the review feedback and then issues one of two decisions: accept the current entity set or return it for revision. If unresolved disputes remain, JdgEn feeds the source text together with targeted review comments back to ExtEn, thereby triggering the next round of focused re-extraction rather than blind regeneration. In this sense, JdgEn functions as the convergence gate of the entity extraction phase. The corresponding prompt templates are shown in Appendix D Fig. 10 (Agent JdgEn Prompt).

4.3.2. Relation extraction phase

Once a stable entity set has been established, Debapt enters the relation extraction phase. This phase extracts semantic relations among the verified entities, which depends not only on local textual evidence, but also on the reliability of the upstream entity results. For this reason, Debapt performs relation extraction only after the entity extraction phase has converged. This phase is carried out by three agents: ExtRe, CritRe, and JdgRe.

ExtRe: relation extraction under schema guidance. Given the verified entity set, ExtRe identifies candidate semantic relations by analyzing contextual content, interaction descriptions, and behavioral dependencies in the source report. Debapt does not treat this phase as open-ended relation discovery. Instead, ExtRe proposes candidate edges under the guidance of the STIX relation schema, which narrows the space of plausible semantic connections and helps preserve the structural coherence of the profile. The corresponding prompt templates are shown in Appendix D Fig. 10 (Agent ExtRe Prompt).

CritRe: relation verification and review. Candidate relations are then examined by CritRe. Specifically, CritRe performs an evidence support check to determine whether each relation is explicitly supported by source evidence, a relation validity check to verify whether its relation type and direction are semantically valid under the permitted STIX vocabulary, a verified entities check to assess whether the connected entity types form a plausible pairing under the current constraints, and a coverage check to examine whether the candidate relations sufficiently cover the verified entities. In addition, CritRe identifies weakly supported, redundant, or logically conflicting candidate edges. This review step filters problematic relations before final acceptance, thereby reducing the risk of introducing semantically invalid edges into the final profile. The corresponding prompt templates are shown in Appendix D Fig. 10 (Agent CritRe Prompt).

JdgRe: adjudication and iterative refinement. JdgRe integrates the current relation proposals and the review feedback to determine whether the relation layer has reached an acceptable state. If the current relation set is sufficiently supported, it is accepted as the final relation output for the current CTI report. Otherwise, the current entity set, the source text, and the review feedback are returned to ExtRe for another round of targeted revision. As in the entity extraction phase, adjudication here determines whether the relation extraction phase should converge or continue. The corresponding prompt templates are shown in Appendix D Fig. 10 (Agent JdgRe Prompt).

4.3.3. Termination mechanism

Because iterative review can improve extraction quality while also increasing computational cost, Debapt introduces an explicit termination mechanism to keep the debate process within reasonable bounds. This mechanism is managed by Mod during phase progression.

After each round, Mod examines whether outstanding disputes have been resolved and whether the latest revision introduces meaningful corrections. A phase terminates under three conditions: when the adjudication result is accepted, when the predefined maximum number of rounds is reached, or when further iterations no longer produce substantive gains. This design allows Debapt to retain sufficient review depth while avoiding ineffective loops caused by repetitive revisions or marginal improvements. In this way, the termination mechanism keeps the debate process controlled and bounded, rather than allowing unnecessary iteration.

4.3.4. Profile aggregation

The outputs of the two debate loops are document-level profiles. To support broader analyst use, Debapt can further aggregate information across multiple CTI reports in a profile aggregation step. Concretely, the threat actor is treated as the central node, and the extracted profile-relevant information from the six VICTOR dimensions is aggregated around it. Each candidate fact is assigned an initial confidence score, while its supporting source documents are indexed in a vector database. During profile aggregation, the framework retrieves semantically related evidence across reports via RAG and evaluates the cross-source consistency of candidate facts. Corroborative or complementary evidence increases the confidence of the corresponding fact, whereas conflicting evidence lowers its confidence and marks it as risky. In this way, scattered document-level profiles can be further synthesized into an actor-centric adversary profile.

5. Evaluation

We conduct extensive experiments to answer the following research questions:

- **RQ1: Overall performance analysis.** Does Debapt outperform competitive supervised and LLM-based baselines in profile-oriented entity and relation extraction for APT adversary profile construction?
- **RQ2: Model variation analysis.** Does Debapt maintain stable performance across different backbone models, and how do model variations affect its effectiveness?
- **RQ3: Ablation study analysis.** How much do the key components in Debapt contribute to the final extraction performance?
- **RQ4: Long-context processing analysis.** How does Debapt behave as CTI reports become longer and more context-intensive?
- **RQ5: In-depth analysis of key mechanisms.** How do Debapt's hallucination mitigation process and Mod-guided termination mechanism support reliable profile construction?

5.1. Experimental setting

To ensure the validity and reproducibility of the Debapt evaluation, we describe the experimental setup, including the datasets, the baseline methods, and the implementation details.

5.1.1. Dataset

We construct the experimental data based on three public datasets. The first is the CTINexus dataset (Cheng et al., 2025), which consists of cybersecurity news briefs and security bulletin summaries. It is used to evaluate the method's ability to identify key security entities and core event clues in relatively short CTI reports. The second is the STIXnet dataset (Marchiori et al., 2023), containing descriptive information about APT groups and threat actors. It assesses the method's capability

to extract profile elements from organization-centered CTI reports. The third is the LLM4CTI dataset (Huang et al., 2026), which comprises complete security technical analysis blogs and threat analysis reports. This dataset evaluates the method's long-context comprehension and cross-sentence semantic association. All three datasets originate from real-world CTI reports and cover a rich variety of security entities, attack behaviors, and semantic associations, providing an experimental foundation for the APT adversary profile construction task.

Since the original annotation objectives, entity category systems, and relation schemas of the datasets do not fully align with the proposed VICTOR ontology, we do not directly adopt their original annotations. Instead, we reconstruct a unified ground truth tailored for the APT adversary profile construction task based on the original data. We adopt the following four-step annotation process: (1) organize the existing triples in the original datasets, map their entities and relations to the type system defined by the VICTOR ontology, and remove elements that do not conform to VICTOR's semantic boundaries; (2) based on the VICTOR ontology, additionally annotate cybersecurity entities and their types that explicitly appear in the text but are not covered by the original annotations; (3) group entities by their types and merge multiple mentions referring to the same concept; and (4) identify the semantic relations between entities and organize them into structured triples in JSON format.

To reduce annotation bias and improve the quality of the ground truth, we adopted an LLM-assisted pre-annotation strategy followed by manual verification. Four graduate students with backgrounds in cyber threat intelligence participated in the annotation process in our research team. Specifically, the first three annotators, including two master's students and one Ph.D. student, independently used the LLM to produce initial silver-standard draft annotations. Based on these drafts, they then conducted item-by-item review, correction, and completion to obtain their individual annotation results. A fourth Ph.D. student served as an arbitrator to resolve disagreements and finalize the gold-standard annotations. To measure the agreement among multiple annotators, we employed Fleiss's kappa (Fleiss, 1971) to evaluate the annotation results. The resulting score is 0.79, indicating substantial agreement. This process yielded 4915 deduplicated independent entities and 5736 relations. To maintain comparability with traditional supervised methods, we further divide each dataset into training, validation, and test sets at a ratio of 6:2:2 and convert them into the BIO annotation format. Supervised baselines utilize the training and validation sets for model learning and parameter selection, while Debapt and other prompt-based methods undergo unified evaluation on the same test set.

5.1.2. Baseline methods

To evaluate the effectiveness of Debapt in extracting threat information for the APT adversary profile construction task, we compare Debapt with both supervised and LLM-based baselines. The supervised baselines include BERT (Xu et al., 2019), SecureBERT (Aghaei et al., 2022), BERT-CRF (Aravind et al., 2023), BERT-BiLSTM-CRF (Dasgupta et al., 2020), BiLSTM (Wang and Liu, 2023), BiLSTM-CRF (Chen et al., 2019), and Extractor (Satvat et al., 2021). These methods represent widely used neural approaches for CTI entity and relation extraction. We also compare Debapt with five LLM-based baselines, including ICL (Dong et al., 2024), CoT (Wei et al., 2022), CTIKG (Huang and Xiao, 2024), CTINexus (Cheng et al., 2025), and AttacKG+ (Zhang et al., 2025). These baselines cover both direct prompting methods and recent CTI-oriented structured extraction frameworks. These comparisons establish the performance baselines and verify the improvements achieved by Debapt.

5.1.3. Implementation and metrics

All experiments in this paper are implemented in a Python 3.11.14 environment. The involved open-source LLMs (e.g., qwen2.5-7b-instruct, qwen2.5-14b-instruct, qwen2.5-32b-instruct, qwen2.5-72b-instruct, qwq-32b, deepseek-r1) are uniformly deployed on Alibaba Cloud Elastic Compute Service (ECS), while the commercial closed-source models (e.g., qwen-turbo, qwen-plus, qwq-plus) are accessed through the official APIs provided by Alibaba Cloud.[2]

Unless otherwise specified, Debapt utilizes the closed-source qwen-plus model as its default thinking-enabled backbone model, with the maximum number of iterations set to 5. To enhance reproducibility and fairness, we uniformly control the key generation parameters of the LLMs. Specifically, the temperature is fixed at 0.2, the maximum output length is set to 8192 tokens, and the random seed is assigned a fixed value. For LLM-based baselines, we provide consistent input contexts, the same VICTOR schema, the same structured output requirements, and the same JSON output format whenever applicable. Since CTINexus relies on exemplar retrieval, we select 10 re-annotated non-test samples from the CTIKG dataset as the exemplar retrieval library.

According to the problem definition in this paper, an APT profile can be represented as a set of structured triples (h, r, t) organized around the threat actor, where h and t denote entity nodes conforming to the constraints of the VICTOR ontology, and r represents the semantic relation between the entities. Therefore, the capability of profile information extraction is primarily reflected in two aspects: (1) whether the entity extraction is accurate and complete; and (2) whether the relation extraction correctly extracts the semantic connections between entities. We evaluate entity extraction and relation extraction separately. We adopt Precision, Recall, and F1-score as evaluation metrics, and further report F1-micro, F1-macro, and F1-weighted to account for differences in report length and information density. The detailed evaluation protocol and metric calculation are provided in Appendix E. To ensure fair comparison, all methods are evaluated using the same matching rules, post-processing procedure, and scoring script. All supervised baselines use the same training and validation sets, while all supervised and LLM-based methods are evaluated on the same test set. Test annotations are not used for model training, parameter tuning, prompt construction, or exemplar selection.

5.2. Overall performance analysis (RQ1)

This experiment aims to evaluate the overall performance of different methods in profile-relevant entity and relation extraction from CTI reports. The capability of profile information extraction is mainly reflected in two aspects: whether the extracted entities are accurate and complete, and whether the extracted relations correctly capture the semantic connections between entities. Tables 1 and 2 report the overall results for entity extraction and relation extraction, respectively. In our reproduction settings for the relevant supervised baselines, we strictly followed the original papers' implementations. Consequently, the derivative methods of BERT and BiLSTM were utilized exclusively for entity recognition and did not include relation extraction modules. We only included Extractor as a comparable supervised baseline in the evaluation of the relation extraction task. Our findings are as follows.

Consistent overall strength. From an overall perspective, Debapt ranks in the top two across all dataset–metric combinations, achieving the best results in 15 of them. This indicates that Debapt maintains highly stable performance across different data distributions and statistical methods. The three datasets originate from CTI reports with varying lengths and information densities; thus, this result demonstrates Debapt's adaptability in both profile information extraction and long-context processing.

Strong and balanced entity extraction. In the entity extraction task, Debapt achieves the best results across all three metrics on both DS1

[2] https://www.aliyun.com/.

Table 1
Overall entity extraction performance on three benchmark datasets. For each column, the bold value indicates the best performance, and the underlined value indicates the second best performance.

Method	DS1(CTINexus)			DS2(STIXnet)			DS3(LLM4CTI)		
	$F1_{micro}$	$F1_{weighted}$	$F1_{macro}$	$F1_{micro}$	$F1_{weighted}$	$F1_{macro}$	$F1_{micro}$	$F1_{weighted}$	$F1_{macro}$
Deep learning-based method									
Bert	0.3378	0.3500	0.3299	0.3574	0.3641	0.4040	0.2623	0.2742	0.2258
SecureBert	0.3007	0.3045	0.2823	0.3386	0.3500	0.3860	0.2470	0.2459	0.2318
Bert_CRF	0.3288	0.3386	0.3133	0.2820	0.1756	0.2522	0.2608	0.2720	0.2364
Bert_bilstm_CRF	0.3201	0.3280	0.2961	0.3117	0.3360	0.3451	0.2444	0.2545	0.2244
Bilstm	0.3359	0.2923	0.3041	0.5011	0.5074	0.5386	0.0481	0.0453	0.0460
Bilstm_CRF	0.5094	0.4787	0.4740	0.5546	0.5494	0.5730	0.0112	0.0111	0.0103
Extractor	0.0665	0.0539	0.0769	0.3281	0.2841	0.2492	0.1276	0.1162	0.1199
LLM-based method									
ICL	0.7306	0.7153	0.7090	0.6944	0.6708	0.7040	0.7242	0.7126	0.6559
CoT	0.7659	0.7409	0.7479	0.6756	0.6568	0.6923	0.7380	0.7285	0.6601
CTIKG	0.6643	0.6424	0.6447	0.7625	0.7513	0.7752	0.6439	0.6188	0.5468
CTINexus	0.7076	0.7240	0.6896	0.7683	0.7571	0.7835	0.6948	0.6663	0.5452
AttacKG+	0.7399	0.7491	0.7195	**0.7839**	0.7024	0.7706	0.6843	0.6938	0.6019
Our method									
Debapt	**0.8213**	**0.8186**	**0.7986**	0.7684	**0.7903**	**0.8193**	**0.8007**	**0.7998**	**0.7243**

Table 2
Overall relation extraction performance on three benchmark datasets. For each column, the bold value indicates the best performance, and the underlined value indicates the second best performance.

Method	DS1(CTINexus)			DS2(STIXnet)			DS3(LLM4CTI)		
	$F1_{micro}$	$F1_{weighted}$	$F1_{macro}$	$F1_{micro}$	$F1_{weighted}$	$F1_{macro}$	$F1_{micro}$	$F1_{weighted}$	$F1_{macro}$
Deep learning-based method									
Extractor	0.0132	0.0118	0.0122	0.0027	0.0242	0.0321	0.0139	0.0143	0.0267
LLM-based method									
ICL	0.7486	0.7296	0.7219	0.7616	0.7438	0.7764	0.8072	0.8052	0.6845
CoT	0.7619	0.7582	0.7479	0.8143	0.8000	0.8187	0.7978	0.7935	0.6281
CTIKG	0.6490	0.6312	0.6020	0.8426	0.8365	0.8645	0.7201	0.7248	0.5840
CTINexus	0.7910	0.8045	0.7642	0.8885	**0.8838**	0.8884	0.8173	0.7976	0.6705
AttacKG+	0.8134	**0.8256**	0.7688	0.8668	0.7686	0.8293	0.8076	0.8031	0.6723
Our method									
Debapt	**0.8197**	0.8199	**0.7944**	**0.8886**	0.8622	**0.8914**	**0.8286**	**0.8440**	**0.7553**

and DS3. On DS2, although its F1-micro (0.7684) is slightly lower than that of AttacKG+ (0.7839), both its F1-weighted (0.7903) and F1-macro (0.8193) are the highest. This suggests that Debapt not only extracts profile-relevant entities effectively, but also maintains more balanced performance across different CTI reports.

Competitive relation extraction. In the relation extraction task, Debapt achieves the best $F1_{micro}$ and $F1_{macro}$ on DS1 and DS2, and the best results on all three metrics on DS3. It is worth noting that on DS2, Debapt's F1-micro of 0.8886 is remarkably close to CTINexus's 0.8885, yet it still secures the highest score. Although its $F1_{weighted}$ on DS2 is lower than that of CTINexus (0.8622 vs. 0.8838), the overall results still show that Debapt is highly competitive in relation extraction.

Clear advantages over supervised methods. Compared with traditional supervised methods, Debapt achieves more effective and stable results across all three datasets. As shown in Table 1, Debapt outperforms all compared supervised methods. Meanwhile, most supervised methods exhibit significant fluctuations across different datasets; particularly on DS3, the entity F1 scores of some methods drop noticeably. In Table 2, Extractor shows overall lower relation extraction performance across the three datasets. A reason for this phenomenon is that traditional supervised methods typically rely heavily on fixed annotation distributions and local feature learning. In contrast, the task in this paper involves cross-sentence and cross-paragraph information integration, as well as more complex semantic mapping within lengthy CTI reports. Consequently, when the dataset distribution changes or the text span increases, their generalization capability is relatively limited.

More robust than single-model LLM baselines. Compared to LLM-based single-model methods, Debapt achieves superior results in most cases, whereas some single-model approaches only perform well on specific datasets. This indicates that the multi-agent debate mechanism is more effective for processing profile-relevant information. Particularly on the F1-macro metric, Debapt gets the highest scores in both entity extraction and relation extraction. This demonstrates that Debapt's multi-agent debate mechanism not only enhances the overall extraction performance but also improves performance consistency across different samples.

5.3. Model variation analysis (RQ2)

This experiment aims to examine whether Debapt remains effective under different backbone models and how model variation affects its profile-oriented information extraction capability. To this end, while maintaining consistent framework settings, prompt constraints, and output formats, we employ various open-source and closed-source large language models, including both thinking-enabled models and non-thinking models, as Debapt's backbone models. We compare the experimental results from the perspectives of both entity extraction and relation extraction to analyze Debapt's model adaptability and transferability. Table 3 presents the results. Overall, while switching models does impact Debapt's performance, the framework maintains

Table 3
Performance of Debapt with different backbone models. For each column, the bold value indicates the best performance, and the underlined value indicates the second best performance.

Model	Entity			Relation		
	$F1_{micro}$	$F1_{weighted}$	$F1_{macro}$	$F1_{micro}$	$F1_{weighted}$	$F1_{macro}$
Open-source models & Non-thinking models						
qwen2.5-7b-instruct	0.6115	0.5581	0.6123	0.7367	0.7222	0.7185
qwen2.5-14b-instruct	0.7514	0.7438	0.7408	0.7591	0.7475	0.7806
qwen2.5-32b-instruct	0.7276	0.6974	0.7461	0.8202	0.8006	0.8241
qwen2.5-72b-instruct	0.7741	0.7545	0.7811	**0.8837**	**0.8666**	**0.8665**
Open-source models & Thinking-enabled models						
qwq-32b	<u>0.8139</u>	<u>0.8004</u>	<u>0.8264</u>	0.8229	0.7910	0.8222
deepseek-r1	0.8125	0.7974	**0.8275**	<u>0.8627</u>	<u>0.8499</u>	<u>0.8653</u>
Closed-source commercial models & Non-thinking models						
qwen-turbo	0.7163	0.6891	0.7145	0.8092	0.7898	0.8211
qwen-plus (non-thinking)	0.7103	0.6817	0.6531	0.7580	0.7343	0.7180
Closed-source commercial models & Thinking-enabled models						
qwq-plus	0.7820	0.7591	0.7906	0.8494	0.8092	0.8384
qwen-plus (thinking-enabled)	**0.8213**	**0.8186**	0.7986	0.8197	0.8199	0.7944

strong adaptability across different types of models. Our findings are as follows:

Different models exhibit different extraction tendencies. From the overall results, no single model dominates across all metrics simultaneously. This indicates that the choice of model affects not only the overall performance but also alters the framework's performance distribution across different subtasks. Some models produce more conservative outputs, while others extract broader sets of profile-relevant facts. This demonstrates that the generation preferences of the models directly influence Debapt's actual extraction behavior.

Thinking-enabled models favor entity extraction. On the entity side, several thinking-enabled models achieve the strongest overall results. For example, qwen-plus (thinking-enabled), qwq-32b, and deepseek-r1 all rank near the top across the three entity metrics. This trend suggests that explicit reasoning can help Debapt extract more complete profile-relevant entities under ontology constraints.

Relation extraction depends on model capability. In the relation extraction task, whether a model is thinking-enabled is not the sole determining factor. Some large-scale open-source non-thinking models can also achieve excellent comprehensive performance. For example, qwen2.5-72b-instruct, which is a non-thinking open-source model, achieves the best relation scores in this table. This indicates that for relation extraction, the model's inherent generation capability and complex semantic modeling ability play equally important roles; a thinking-enabled setting does not necessarily confer a consistent advantage.

Larger-scale models tend to perform better. Within the Qwen 2.5 open-source model series, the performance of profile information extraction exhibits an upward trend as the parameter size increases, especially when moving from 7B and 14B models to the 72B model. This phenomenon indicates that a larger parameter scale generally helps enhance the model's ability to integrate long-context information and provides an advantage in modeling complex semantic relations.

Robust adaptability across model sources. The comparison between open-source models and closed-source commercial models reveals that closed-source models do not consistently dominate across all metrics; open-source models can also achieve competitive results under various settings. This indicates that the effectiveness of Debapt comes mainly from its ontology-guided debate design rather than from dependence on one particular model source, demonstrating robust model compatibility.

5.4. Ablation study analysis (RQ3)

This experiment aims to examine how the key agents and designs in Debapt contribute to the final extraction performance. To this end, through an ablation study, we remove four components in turn: the team-based extraction design in the entity extraction phase, the entity review agent, the relation review agent, and the adjudication agents. We then compare the performance changes of each variant in both the entity and relation extraction tasks to evaluate the specific roles of different components within the framework. As shown in Table 4, removing any core component leads to a decline in performance. Different agents contribute in different ways, and their effects are not equally distributed across the entity extraction phase and relation extraction phase. Specifically:

The team-based entity design is important for the entity extraction phase. Removing ExtEn-Team causes a clear drop in entity performance, with E-$F1_{micro}$ and E-$F1_{macro}$ decreasing by 14.7% and 13.7%, respectively. In contrast, the relation scores decrease only slightly. This result suggests that the team-based ontology-aligned extraction design mainly strengthens the quality of the entity layer, while its indirect effect on the relation layer is more limited.

CritEn is the most influential component. Removing the CritEn agent results in the most pronounced degradation in both entity and relation extraction. Specifically, the entity side experiences the largest decline, and the relation side also shows a substantial drop. This indicates that the entity review not only determines the quality of the entity extraction phase but also serves as a crucial prerequisite safeguard for the relation extraction phase. Overall, CritEn is the most impactful component in the framework.

CritRe mainly improves relation quality. Removing CritRe results in only a slight decline in the entity metrics, whereas the decrease in the relation metrics is significantly more pronounced. Specifically, the R-$F1_{micro}$ score drops by 11.6%, marking the largest decline among the relation micro-average metrics. This result indicates that relation review plays a focused role in checking, filtering, and refining relation level outputs.

Adjudication stabilizes iterative refinement. Removing JdgEn and JdgRe also causes a clear performance drop in both phases. This result suggests that adjudication is important for dispute resolution and phase convergence. Without this mechanism, the debate process becomes less stable, and the final outputs become less reliable.

Overall, the removal of components related to the entity extraction phase generally triggers a broader cascading effect on the overall

Table 4
Ablation study of Debapt after removing key agents and designs. Bold values indicate the performance of the full model. E and R denote entity and relation, respectively, and Δ denotes the relative performance change compared with the full model. w/o ExtEn-Team removes the team-based extraction design in the entity extraction phase, w/o CritEn removes the entity review agent, w/o CritRe removes the relation review agent, and w/o JdgEn & JdgRe removes the entity and relation adjudication agents.

Variant	E-F1$_{micro}$	E-F1$_{macro}$	R-F1$_{micro}$	R-F1$_{macro}$	ΔE-F1$_{micro}$	ΔE-F1$_{macro}$	ΔR-F1$_{micro}$	ΔR-F1$_{macro}$
Ours	**0.8213**	**0.7986**	**0.8197**	**0.7944**	–	–	–	–
w/o ExtEn-Team	0.7008	0.6888	0.7969	0.7927	−14.7%	−13.7%	−2.8%	−0.2%
w/o CritEn	0.5897	0.5398	0.7435	0.6450	−28.2%	−32.4%	−9.3%	−18.8%
w/o CritRe	0.7881	0.7947	0.7245	0.6954	−4.0%	−0.5%	−11.6%	−12.5%
w/o JdgEn & JdgRe	0.6282	0.5734	0.7592	0.7048	−23.5%	−28.2%	−7.4%	−11.3%

Table 5
Statistics of CTI report groups with different average token lengths.

Group	Min tokens	Max tokens	Avg tokens
avg150	129	160	150.75
avg300	312	350	324.50
avg500	479	603	532.00
avg1000	1058	1335	1217.33
avg2000	1736	2458	2032.67

performance, among which CritEn exhibits the most significant impact. Meanwhile, the influence of CritRe is more concentrated within the relation extraction phase. This observation is consistent with the functional division of the respective components within the framework.

5.5. Long-context processing analysis (RQ4)

This experiment aims to analyze how Debapt behaves as CTI reports become longer and more context-intensive. Since this analysis focuses on long-context robustness, we use ICL and CoT as the comparative baselines. We divide the test reports into five groups according to report length. Table 5 reports the minimum, maximum, and average token counts of each group. We then compare the F1-micro scores of entity and relation extraction across the three methods. The results are shown in Fig. 4. Overall, Debapt remains more stable than ICL and CoT as report length increases. The two single-model prompting baselines show a clearer downward trend on longer reports, while Debapt fluctuates much less. This pattern suggests that Debapt is better able to maintain extraction quality when information is distributed across longer contexts.

For entity extraction, Debapt maintains consistently high F1-micro scores across all five length groups. By contrast, the performance of ICL and CoT decreases more noticeably as report length grows, especially in the longer groups. The gap curves in Fig. 4(a) also show that Debapt's advantage over both baselines becomes larger on longer reports.

A similar pattern appears in relation extraction. ICL and CoT still remain competitive on shorter and medium-length reports, but their performance drops more clearly in the longer groups. Debapt, in contrast, remains comparatively stable, and its advantage becomes more visible as report length increases. This trend is especially clear in the avg1000 and avg2000 groups.

These results suggest that Debapt is more robust than ICL and CoT under long-context conditions. A likely reason is that Debapt does not rely on single-round prompting alone. Instead, it performs staged profile information extraction through ontology-guided debate, iterative review, and adjudication. This design helps preserve profile-relevant information and reduces the information omission and performance degradation caused by context growth.

5.6. In-depth analysis of key mechanisms (RQ5)

This experiment investigates two key mechanisms in Debapt that are designed to improve the reliability of profile construction. Specifically, we analyze the hallucination mitigation process to examine whether iterative debate can effectively reduce unsupported or incorrect outputs, and we further study the Mod-guided termination mechanism to evaluate how the framework controls execution boundaries and avoids ineffective iterations during multi-round refinement.

5.6.1. Analysis of hallucination mitigation

This experiment aims to analyze how hallucination changes as the number of debate rounds increases. Fig. 5 reports both the hallucination ratio and the hallucination count across rounds. Overall, both the hallucination ratio and the hallucination count decrease steadily across rounds, indicating that the multi-round review and feedback mechanism is effective in suppressing hallucinated outputs.

More specifically, the hallucination count decreases from 107 in R1 to 13 in R5, corresponding to an overall reduction of approximately 87.9%. Similarly, the hallucination ratio drops from 0.1080 to 0.0124, with an overall reduction of about 88.5%. The largest decrease occurs in the early rounds, from R1 to R3, which suggests that the first few rounds play the most important role in correcting unsupported generations and filtering unreliable outputs. From R3 to R5, both metrics continue to decline, but the improvement becomes noticeably smaller, indicating that the results gradually converge.

Furthermore, to illustrate how Debapt reduces hallucinations and improves extraction quality, we present two representative examples of the CritEn and CritRe processing procedures (Fig. 6).

CritEn (entity-level review). In an APT28-related excerpt, the initial entity list produced by ExtEn contains three typical errors. First, there is a type mismatch: "Sourface" is mistakenly labeled as threat_actor_alias. Second, there is a verbatim inconsistency: "Fancy Bear" is extracted as "FancyBear". Third, one alias, "Sofacy", is omitted. Following strict grounding and exact-string requirements, CritEn flags "Sourface" as not fitting any Originator (O) field and recommends removing it from the O-dimension list, corrects "FancyBear" to the verbatim form "Fancy Bear" and changes its type to threat_actor_alias, and proposes adding the missing alias "Sofacy". These revisions are then applied in the next extraction round to obtain an O-dimension entity set that is both more complete and schema-consistent.

CritRe (relation-level review). We further show a CVE-centric excerpt involving CVE-2023-46604, Apache ActiveMQ server, uFSyL-szKsuR.exe, and 166.62.100[.]52. ExtRe outputs triples that violate hard constraints, including a disallowed relation name (e.g., execute) and a non-verbatim endpoint (e.g., 166.62.100.52 does not match the verified entity string). CritRe first adds triples to ensure that they sufficiently cover the verified entities (e.g., Apache ActiveMQ server), then deletes invalid triples and adds compliant replacements using only allowed relation names and verified entity strings (e.g., correcting the communication endpoint to 166.62.100[.]52).

These results show that Debapt's iterative debate mechanism is effective for hallucination mitigation and helps produce more reliable profile-relevant facts.

5.6.2. Analysis of Mod-guided termination mechanism

To examine how Mod controls the execution boundary of Debapt and prevents unproductive iterations, we conduct a small case study using five real CTI report excerpts. Each excerpt is a contiguous and

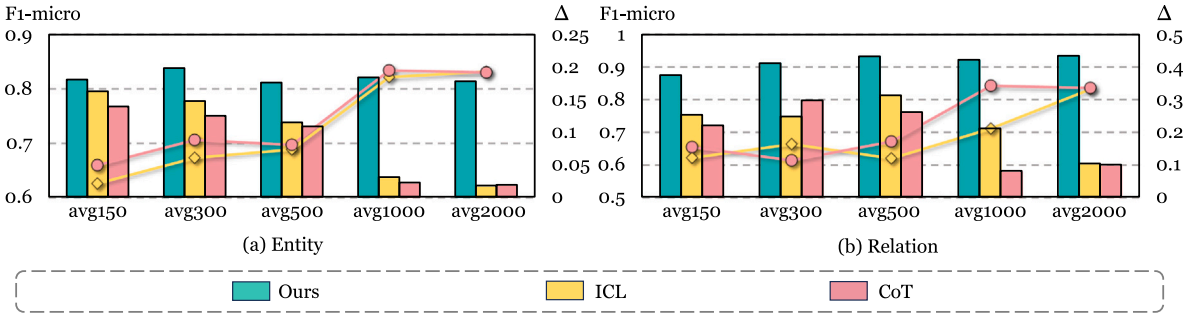


Fig. 4. Effect of report length on the performance of Debapt. The *x*-axis denotes report groups with different average token lengths, where tokens are counted with the cl100k_base tokenizer (tiktoken). The bar charts report the F1-micro scores of different methods, corresponding to the left *y*-axis. The line charts report the F1-micro gap between Debapt and the baselines, corresponding to the right *y*-axis. The yellow line denotes the gap between Debapt and ICL, and the red line denotes the gap between Debapt and CoT. Subfigure (a) reports entity extraction results, and subfigure (b) reports relation extraction results.

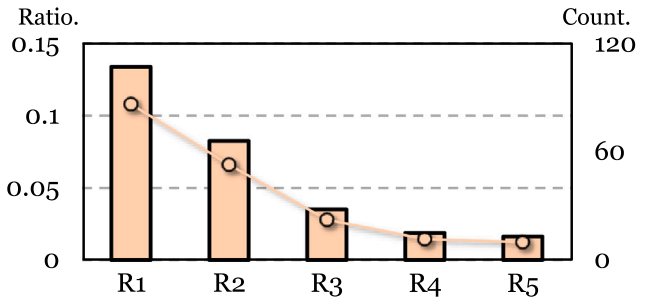


Fig. 5. Effect of iteration rounds on hallucination mitigation. The *x*-axis denotes the iteration round. The line denotes the hallucination ratio at each round, and the bars denote the corresponding hallucination count. The left *y*-axis corresponds to the ratio, and the right *y*-axis corresponds to the count.

Table 6
Termination outcomes of Debapt on five representative CTI report excerpts. For each excerpt, we report the number of debate rounds and the final termination status in the entity extraction phase and relation extraction phase. In the entity extraction phase, Avg. Rounds averages the rounds over six ontology-aligned dimensions, while Rounds in the relation extraction phase counts relation refinement rounds. Final Status records Mod's stopping decision (Converged means the phase reaches a stable state where further iterations are not necessary, while MaxRounds means the phase stops after reaching the preset maximum number of rounds).

Report	Entity extraction phase		Relation extraction phase	
	Avg. Rounds	Final status	Rounds	Final status
CozyDuke	3	Converged	5	MaxRounds
Carbanak_APT	5	MaxRounds	5	MaxRounds
FSOFACY	3	Converged	4	Converged
Kimsuky	4	Converged	5	MaxRounds
apt28	4	Converged	5	Converged

representative part of the original report, with an average length of about 3000 tokens. We run the full Debapt pipeline and record, for each report, the number of debate rounds and the final termination decision in both phases, as shown in Table 6.

As shown in Table 6, the entity extraction phase usually converges within a small number of rounds. Four of the five reports terminate with "Converged", and only one reaches "MaxRounds". This suggests that the extract–review–adjudicate loop guided by JdgEn can usually stabilize the entity set efficiently. In contrast, the relation extraction phase is more demanding. Two reports terminate with "Converged", while the remaining three stop at "MaxRounds". This pattern suggests that relation extraction is harder under long-form CTI narratives, where semantic links often depend on more dispersed evidence and more complex reasoning.

In such cases, Mod provides an effective stopping safeguard. When additional rounds produce only limited changes or unresolved disputes remain after repeated refinement, Mod terminates the phase according to the preset condition. This mechanism keeps inference cost under control and prevents ineffective iterations from extending the workflow without meaningful gains.

5.7. Case study

To illustrate the utility of profile aggregation in Debapt, we conduct a case study on two representative APT groups, Lazarus Group and APT28. For each group, we aggregate the extracted profile-relevant information from multiple CTI reports and generate an actor-centric adversary profile card. Each card contains three parts: a summary header (attribution, aliases, and corpus scale), a VICTOR coverage table (top-3 items for each field), and chart-based summaries of major trends. This design allows analysts to grasp the overall profile of a threat actor without repeatedly reading the full set of raw reports.

Fig. 8 shows the generated adversary profile card for Lazarus Group, including aliases such as Hidden Cobra, APT38, and ZINC. We process 25 CTI reports that span the period from 2007 to 2024. Debapt extracts 3940 triples and 4310 entity mentions across the six VICTOR dimensions. The resulting profile reveals dual motivations of espionage and financial gain. The charts show clear behavioral signatures. The group predominantly uses phishing for initial access, employs LOLBins (e.g., cmd.exe and rundll32.exe), and relies on malware families such as TangoBravo and TangoCharlie. The target distribution is broad. The Victim Location field shows South Korea as the top target, followed by a long tail across East Asia, the Middle East, Europe, and North America. This suggests a global operating scope rather than a single-region focus. Government appears as the main sector, while other sectors also appear, indicating diverse targeting. Furthermore, ATT&CK mapping indicates comprehensive end-to-end intrusion capabilities using HTTP/HTTPS C2 channels. Key techniques (e.g., T1543.003, T1071.001, and T1082) reflect a standard operational sequence of initial delivery, persistence, C2 communication, and execution. The activity timeline shows continuous activity, especially with prominent peaks between 2019 and 2021 aligning with major campaigns such as AppleJeus and ThreatNeedle.

Additionally, Fig. 9 shows the generated adversary profile card for APT28, including aliases such as Sednit, Fancy Bear, Sofacy, and STRONTIUM. We process 22 CTI reports and extract 2061 triples and 2179 entity mentions. The generated profile indicates a clear espionage

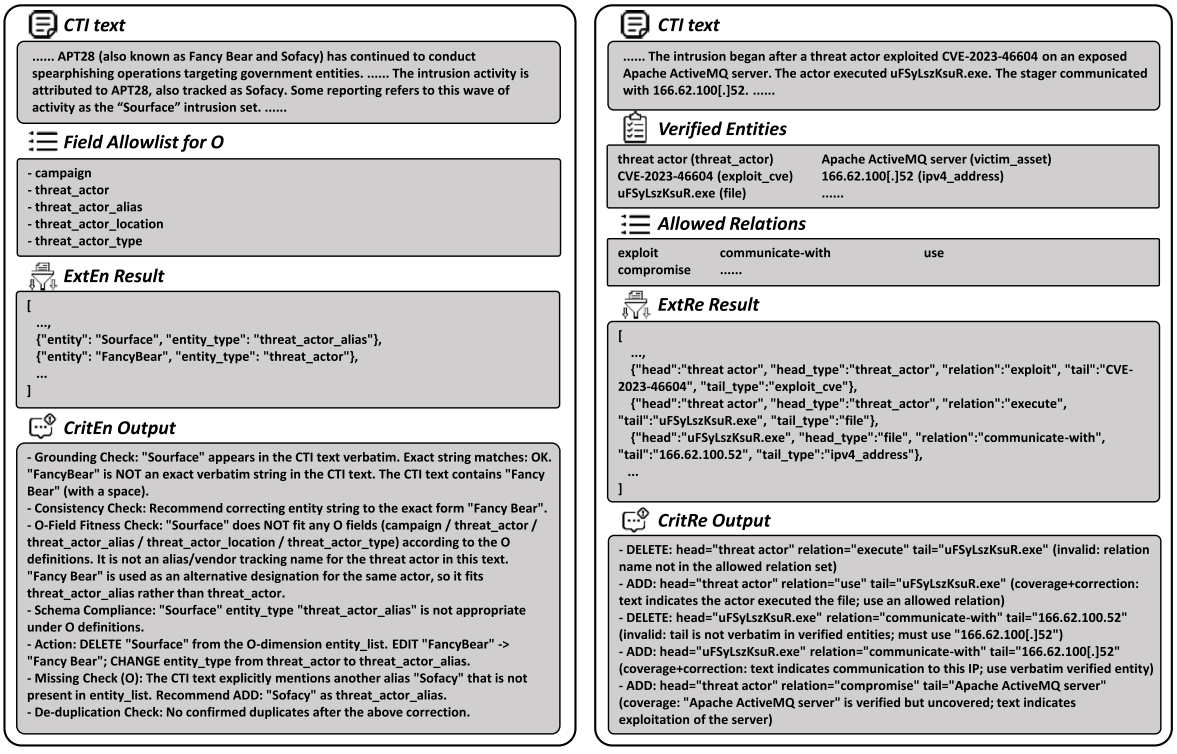


Fig. 6. Representative examples of CritEn (Left) and CritRe (Right) correcting entity-level and relation-level errors during iterative debate.

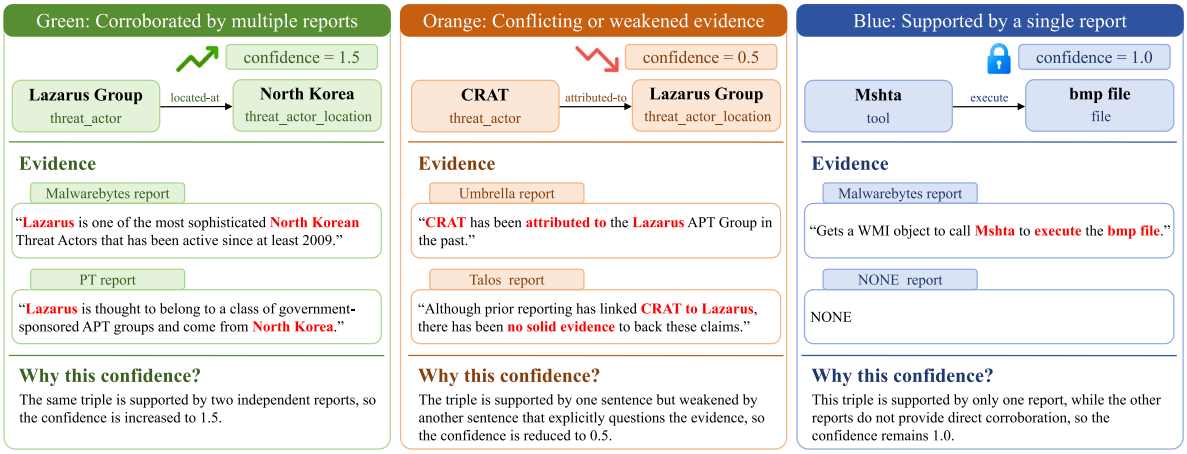


Fig. 7. Cases of confidence scoring for representative profile facts under multi-source evidence aggregation. Green indicates facts corroborated by multiple reports, blue indicates facts supported by a single report, and orange indicates facts weakened by conflicting evidence.

motivation focused on credential theft. Behaviorally, APT28 prioritizes phishing and zero-day exploits, relying on a concentrated malware ecosystem (e.g., Sofacy and CHOPSTICK) and diverse C2 protocols (HTTP, SMTP, and DNS). Target analysis reveals a specific focus on government, military, and defense industry sectors across Ukraine, the United States, and Georgia. The tactics and techniques span multiple phases and reflect a systematic workflow from reconnaissance to execution. Notably, compared to Lazarus Group, APT28 exhibits broader vulnerability exploitation, prominently featuring such as CVE-2015-1701 and CVE-2015-2590 within a systematic attack workflow. The activity timeline has a clear peak in 2014–2016, matching major public cases such as the DNC intrusion, the attack on the German Bundestag, and broad targeting of Eastern European government networks. The bars drop in 2017–2019, which is consistent with a period of reduced visible activity after broad public attribution and indictments. This may reflect stricter operational security.

To further strengthen the case study, we additionally examine how Debapt assigns confidence scores to representative profile facts under multi-source evidence aggregation. As illustrated in Fig. 7, facts corroborated by multiple independent reports receive an increased confidence score of 1.5, facts supported by a single report retain the default confidence score of 1.0, and facts weakened by conflicting evidence receive a reduced confidence score of 0.5. Together with the supporting evidence snippets, this design helps analysts quickly assess the reliability and traceability of aggregated profile facts.

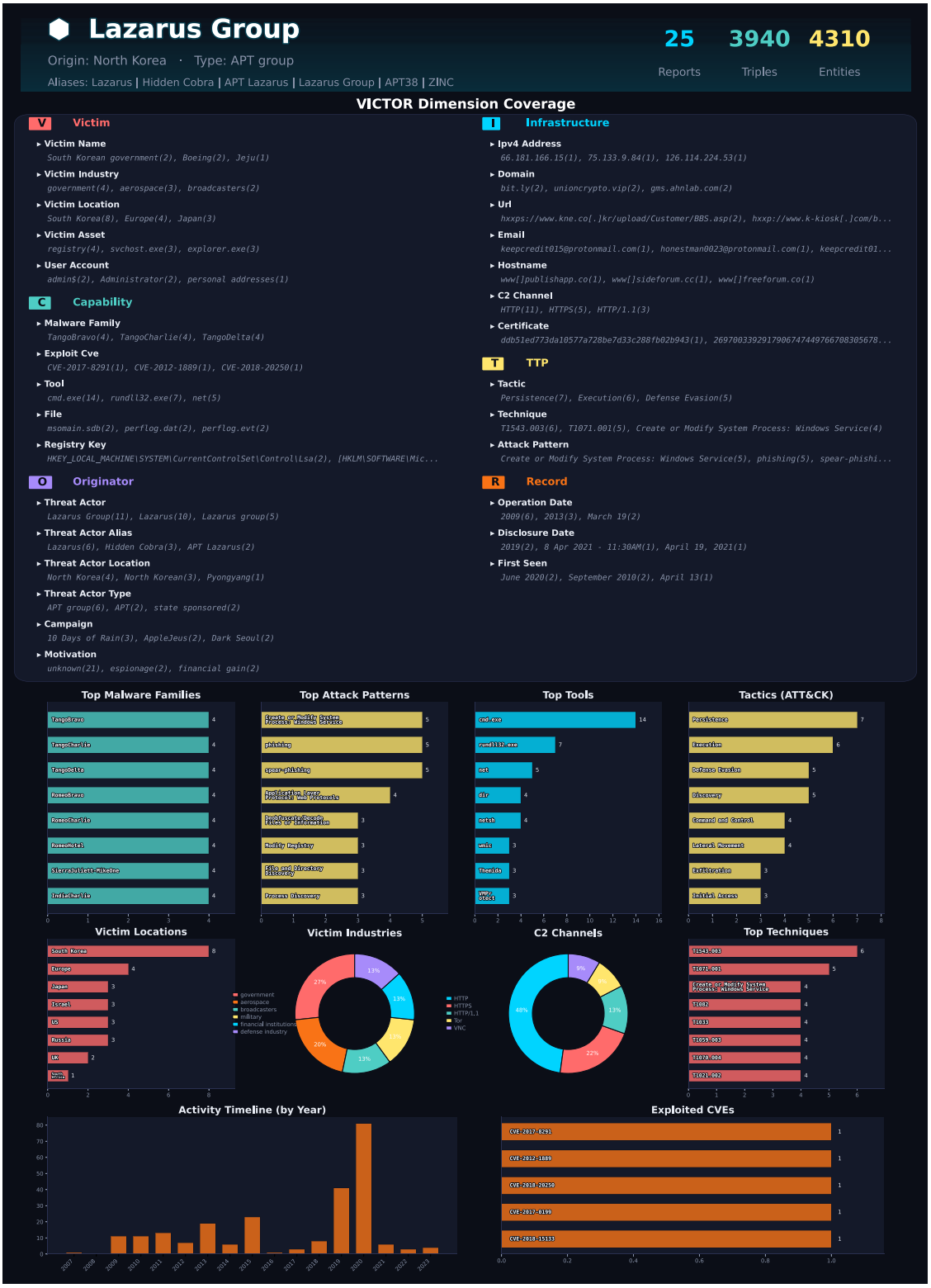


Fig. 8. Actor-centric adversary profile of Lazarus Group.

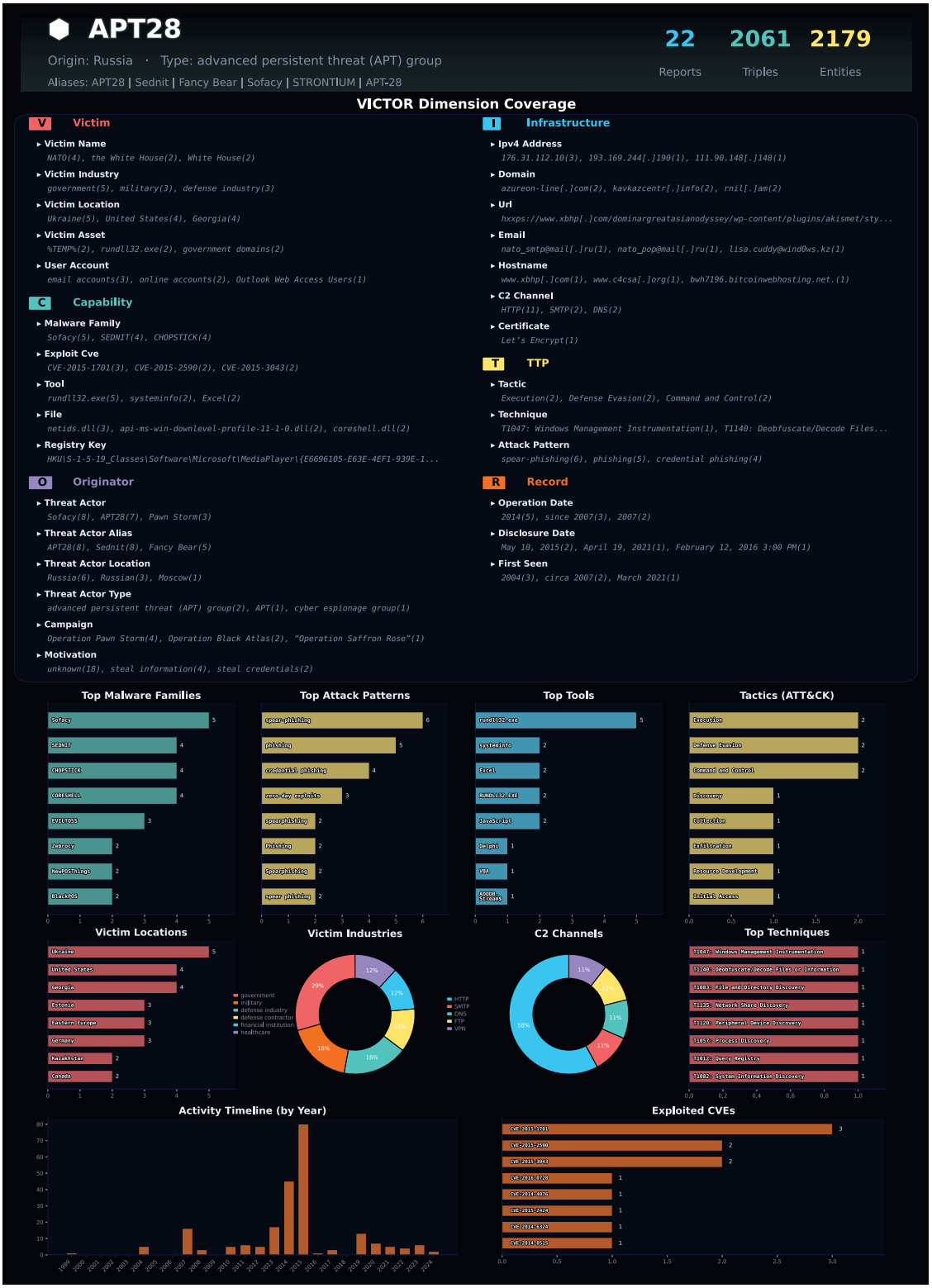


Fig. 9. Actor-centric adversary profile of APT28.

Overall, the Debapt-generated adversary profile cards combine structured fields with chart-based summaries and provide a compact view of actor-centric. They reduce the effort required to synthesize multiple reports and support practical adversary profiling and cross-actor comparison.

6. Conclusion

In this work, we study APT adversary profile construction from CTI reports. We formulate this task as an ontology-constrained profile information extraction task. The goal is to turn fragmented CTI information into structured and reliable adversary profiles.

To support this task, we develop the VICTOR ontology and propose Debapt, an ontology-driven multi-agent debate framework. Without task-specific fine-tuning, Debapt performs adversary profiling through two debate loops in the entity extraction phase and relation extraction phase. Guided by ontology constraints and moderator-controlled debate, the framework improves stability and schema compliance during profile information extraction.

Experiments on three re-annotated CTI datasets show that Debapt improves profile-oriented entity and relation extraction over competitive baselines. Additional analyses further show its robustness on long CTI reports and the value of its key mechanisms for hallucination mitigation and controlled convergence. Case studies show that the extracted profile-relevant information can support the construction of analytically useful adversary profiles.

Future work will extend this line of research to more complex intelligence settings. One important direction is multi-actor profiling, since some CTI reports, particularly those organized around shared TTPs, may describe multiple APT groups within a single narrative. Extending the framework to maintain and distinguish multiple concurrent profile states in such cases would support finer-grained attribution.

Author contributions

Xinyun Zhao and Lanlan Qi contributed equally to this work. The manuscript was written through the contributions of all authors. All authors have approved the final version of the manuscript.

CRediT authorship contribution statement

Xinyun Zhao: Writing – original draft, Software, Methodology, Data curation, Conceptualization. **Lanlan Qi:** Writing – review & editing, Supervision, Formal analysis, Conceptualization. **Yongheng Zhang:** Writing – review & editing, Visualization, Investigation, Data curation. **Yingxiao Guan:** Writing – review & editing, Visualization, Investigation, Data curation. **Guozheng Yang:** Writing – review & editing, Supervision. **Yuliang Lu:** Writing – review & editing, Supervision. **Xiang Wang:** Writing – review & editing, Supervision.

Declaration of generative AI in scientific writing

Generative AI and AI-assisted technologies should only be used in the writing process to improve the readability and language of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Unified VICTOR field definitions

Appendix A provides detailed information on VICTOR to complement the conceptual ontology in Section 4. While the main text introduces the six profile dimensions at a high level, Table 7 defines the extraction boundary and semantic granularity of each field in a unified format. These definitions serve as shared constraints for role prompts in Debapt, ensuring consistent handling of heterogeneous evidence in CTI reports.

Appendix B. Standard alignment of VICTOR

Appendix B provides a alignment between VICTOR and representative STIX or MITRE ATT&CK concepts. This alignment is intended to clarify how VICTOR inherits standard CTI semantics while adding profiling-specific fields required for APT adversary profile construction. As shown in Table 8, several VICTOR fields can be directly represented by existing STIX objects or ATT&CK concepts, such as IP addresses, malware, tools, vulnerabilities, techniques, and campaigns. Meanwhile, some fields are introduced to support actor-centric profiling and do not always have a single fixed counterpart in existing standards. For example, `victim_industry`, `hostname`, and `operation_date` may need to be interpreted through multiple STIX objects, attributes, or ATT&CK concepts depending on context. Therefore, the table reports both direct alignments (D) and semantic alignments (S) to reflect the compatibility and profiling-oriented nature of VICTOR.

Appendix C. Relation semantic guidance of VICTOR

Appendix C provides the type-level relation semantic guidance used in VICTOR. To make the relation hints compact and readable, we first define several type-set abbreviations in Table 9. These abbreviations group frequently used VICTOR entity types. Based on these type sets, Table 11 summarizes the possible source and target entity types for each relation. These type-level hints are derived from STIX-compatible relation semantics and APT adversary profiling needs. They are used as lightweight semantic guidance during relation extraction and review, helping agents identify semantically reasonable relation candidates. The guidance is not used as a rigid graph constraint. A relation can still be retained when it is well supported by the source text and semantically appropriate in context.

Appendix D. Agent prompts of Debapt

Appendix D presents the operational prompt templates (Fig. 10) that instantiate the Debapt framework described in Section 4. Beyond process descriptions, these prompts explicitly define role responsibilities, required output structures, and ontology-constrained checks for extraction, review, and adjudication under moderator coordination. By requiring evidence-grounded review, exact-string consistency, and schema-compliant relation constraints, the templates convert VICTOR semantics into executable instructions and reduce ambiguity across debate rounds. This appendix complements the quantitative and case-study results in Section 5.

Appendix E. Evaluation protocol and metric

Appendix E provides the detailed matching rules and metric calculation formulas used for entity extraction and relation extraction evaluation.

For each document, we extract entities from the head and tail fields of all triples and remove duplicate entity mentions. Entity matching is performed in two steps. First, we apply case-insensitive exact string matching. If exact matching fails, we apply a weak semantic matching step to handle equivalent entity expressions, such as

Table 7
Unified field definitions of the VICTOR ontology.

Dimension	Field	Definition
Victim (V)	`victim_name`	Name of the victim organization or entity explicitly identified as the target.
	`victim_industry`	Victim industry sector, such as finance, government.
	`victim_location`	Victim geographic location, such as country, region, or city.
	`victim_asset`	Victim-side asset or system explicitly reported as compromised or impacted.
	`user_account`	Account explicitly targeted or abused during the attack.
Infrastructure (I)	`ipv4_address`	IPv4 address used, controlled, or referenced in attacker infrastructure.
	`ipv6_address`	IPv6 address used, controlled, or referenced in attacker infrastructure.
	`mac_address`	MAC address identifying a device interface involved in attacker infrastructure.
	`domain`	Attacker-registered or attacker-abused FQDN used for delivery, redirection, or C2.
	`email`	Email address used for phishing, social engineering, or C2-related contact.
	`url`	Full URL used for redirection, delivery, or hosting, including protocol and path.
	`certificate`	TLS/SSL certificate attribute explicitly identifying infrastructure.
	`c2_channel`	Explicit command-and-control communication channel or platform (e.g., DNS).
	`hostname`	Hostname or node name explicitly identifying an infrastructure host, or server.
Capability (C)	`malware_family`	Malware family name or specific malware or implant name.
	`exploit_cve`	CVE identifier mentioned as exploitation-relevant (e.g., CVE-2023-23397).
	`tool`	Tool or framework used by the attacker, including legitimate or dual-use tools.
	`file`	File-level artifact tied to attacker capability (e.g., payload or script name).
	`registry_key`	Windows registry key or path created, modified, or abused as part of the attack.
TTPs (T)	`tactic`	ATT&CK tactic tag (e.g., Defense Evasion or TA0005).
	`technique`	ATT&CK technique or sub-technique tag (e.g., T1078, Phishing, or T1542.003).
	`attack_pattern`	Attack pattern or method describing how the attacker conducts the intrusion.
Originator (O)	`campaign`	Name of a coordinated attack campaign.
	`threat_actor`	Name of the threat actor or group explicitly identified as conducting the attack.
	`threat_actor_alias`	Alias or vendor tracking name mapped to the same threat actor.
	`threat_actor_location`	Geographic origin or background of the threat actor.
	`threat_actor_type`	Threat actor category (e.g., cybercrime group, or espionage group).
Record (R)	`operation_date`	Date, time, or time range when attacker operations or activities occurred.
	`disclosure_date`	Date or time when the incident or activity was first publicly disclosed or reported.
	`first_seen`	Date or time when the activity, threat actor, or campaign was first observed.

Table 8
Alignment between VICTOR fields and representative STIX or MITRE ATT&CK concepts. D denotes direct alignment, where VICTOR fields have direct or clear attribute-level counterparts in existing standards; S denotes semantic alignment, where VICTOR fields are profiling-oriented semantics that may be represented through multiple STIX objects, attributes, or ATT&CK concepts without a single fixed counterpart.

Dimension	VICTOR fields	Aligned standards (STIX and MITRE)	Type
Victim (V)	`victim_name`, `victim_industry`, `victim_location`, `victim_asset`, `user_account`	**STIX** *Identity*, *Location*, *Infrastructure*, *Software*, and *User Account*	D/S
Infrastructure (I)	`ipv4_address`, `ipv6_address`, `mac_address`, `domain`, `email`, `url`, `certificate`, `c2_channel`, `hostname`	**STIX** *IPv4 Address*, *IPv6 Address*, *MAC Address*, *Domain Name*, *Email Address*, *URL*, *X.509 Certificate*, *Network Traffic*, and *Infrastructure* **MITRE** *Command and Control related concepts*	D/S
Capability (C)	`malware_family`, `exploit_cve`, `tool`, `file`, `registry_key`	**STIX** *Malware*, *Vulnerability*, *Tool*, *File*, *Artifact*, and *Registry*	D
TTPs (T)	`tactic`, `technique`, `attack_pattern`	**MITRE** *Tactic*, *Technique*, and *Sub-technique* **STIX** *Attack Pattern*	D
Originator (O)	`campaign`, `threat_actor`, `threat_actor_alias`, `threat_actor_location`, `threat_actor_type`	**STIX** *Campaign*, *Threat Actor*, *Location*, and *Alias*	D/S
Record (R)	`operation_date`, `disclosure_date`, `first_seen`	**STIX** *Date*	D/S

Table 9
Type-set abbreviations used in the relation semantic guidance of VICTOR.

Symbol	Entity types
$\mathcal{I}_{net}$	{`ipv4_address`, `ipv6_address`, `domain`, `url`, `hostname`}
$\mathcal{I}_{c2}$	{`ipv4_address`, `ipv6_address`, `domain`, `url`, `hostname`, `c2_channel`}
$\mathcal{I}_{infra}$	{`ipv4_address`, `ipv6_address`, `domain`, `url`, `hostname`, `email`, `certificate`, `c2_channel`, `mac_address`}
$\mathcal{A}_{actor}$	{`campaign`, `threat_actor`}
$\mathcal{A}_{action}$	{`attack_pattern`, `campaign`, `malware_family`, `threat_actor`, `tool`}

Agent ExtEn Prompt

```
[Task]
You are X-ExtEn (X Dimension Entity Extractor).
Given the CTI text below, extract ALL X-related entities strictly according to the field definitions and field_allowlist.  // If round = 0
Re-extract X-dimension entities using the reviewer feedback to correct mistakes, while ensuring all DISTINCT X-related entities in the text are captured.  // If round > 0

[field_allowlist and definitions for X]{field_allowlist}

[Hard Rules]
1. COMPLETE COVERAGE: Extract every X-related entity in the CTI text. Do NOT omit any.
2. STRICT GROUNDING: Only extract / keep or add entities that appear verbatim in the CTI text. Do NOT infer, guess, or expand.
3. // If round > 0:
   USE REVIEW (with verification): Verify each reviewer suggestion against the CTI text before applying it. Apply only if supported by the text and consistent with the field definitions:
   - Entity absent from text → REMOVE (only after confirming absence).
   - String mismatch → EDIT to verbatim form in text.
   - Wrong field assignment → CHANGE entity_type within field_allowlist.
   - Missing entity suggested → ADD only if verbatim in text and truly distinct.
   - Entity fits no field definition → REMOVE even if it appears in text.
4. STRICT SCHEMA: entity_type MUST be one of the field_allowlist values.
5. NO DUPLICATES: Output each distinct entity once; prefer the most complete verbatim form.
6. NO EXTRA FIELDS: Output ONLY the entity_list. No explanations.
7. If NO X-related entity exists, output exactly: "no entity".

[Output Format]
[{"entity": "<verbatim string from text>", "entity_type": "<field_allowlist value>"},...]

[CTI Text]{CTI_TEXT}
[Previous entity_list]{PREV_ENTITY_LIST}  // If round > 0, append
[Previous round review] {PREV_REVIEW}  // If round > 0, append
```

Agent ExtRe Prompt

```
[Task]
You are ExtRe (Relation Identifier).
Given the CTI text and verified entities, identify ALL relation triples between these entities that are clearly supported by the CTI text.  // If round = 0
Re-extract and correct relation triples using the reviewer feedback. Keep what is correct, remove what is wrong, and add what is clearly missing.  // If round > 0

[Allowed Relation Names]{RELATION_NAME_SET}
[Relation Type Hints (optional)]{RELATION_TYPE_HINTS}

[Hard Rules]
1. VERIFIED ENTITIES ONLY: Every head and tail MUST be a verbatim entity string from the verified entities list. Do NOT introduce new entities.
2. ALLOWED RELATIONS ONLY: relation MUST be exactly one of the allowed relation names.
3. CTI-BASED LINK FIRST, LABEL SECOND: Output a triple ONLY when the CTI text clearly indicates a relationship between head and tail. Co-mention or proximity alone is NOT sufficient. After confirming the link, choose the single best-fitting allowed relation label.
4. TYPE HINTS ARE OPTIONAL: A triple is valid if the CTI text supports it and the relation name is allowed, even if head/tail types deviate from the hints.
5. // If round > 0:
   VERIFY BEFORE APPLYING FEEDBACK: Apply ADD/DELETE/CHANGE/REVERSE only if the suggestion is supported by the CTI text and consistent with these rules. If feedback is unsupported or overreaches, ignore it and treat the CTI text as source of truth.
6. NO DUPLICATES: Output each (head, relation, tail) triple only once.

[Output Format]
[{"head": "<verbatim entity>", "relation": "<allowed relation>", "tail": "<verbatim entity>"},...]

[CTI Text]{CTI_TEXT}
[Verified Entities]{VERIFIED_ENTITIES}
[Previous triples]{PREV_TRIPLES}  // If round > 0, append
[Reviewer feedback]{PREV_REVIEW}  // If round > 0, append
```

Agent CritEn Prompt

```
[Task]
You are X-CritEn (X Dimension Entity Reviewer).
Audit whether CTI text truly contains NO X-related entities.  // If entity_list = "no entity" or empty.
Review the extracted entity_list against the CTI text and the field definitions. Identify all errors and provide actionable correction suggestions.  // Else

[field_allowlist and definitions for X]{field_allowlist}

[Hard Rules]
1. STRICT GROUNDING: Only judge using what is explicitly present in CTI text. Do NOT infer.
2. COMPLETE SCAN: Scan the entire CTI text to check for any X-related entities.
3. OUTPUT ONLY REVIEW: Output ONLY a plain-text review string. No JSON, no extra keys.

[Review Checklist]  // Applied only when entity_list is not "no entity"
1. For each extracted item:
- Grounding: Does the entity appear verbatim in the CTI text?
- Exactness: Does the extracted string exactly match the text character-by-character?
- Field Fitness: Does the entity fit any field in field_allowlist per its definition?
- Schema: Is entity_type the correct field for this entity?
2. Missing Check:
- Identify any X-related entities present in the CTI text but absent from entity_list.
- Before recommending ADD, confirm the candidate is truly distinct from existing items.
- For each truly missing entity: provide the verbatim string and correct entity_type.
3. De-duplication Check:
- Recommend MERGE only when two items provably refer to the same entity.

[CTI Text]{CTI_TEXT}
[entity_list]{entity_list}
[Output Requirement] Output ONLY the review string with concrete, executable suggestions.
```

Agent CritRe Prompt

```
[Task]
You are CritRe (Relation Reviewer).
Audit whether the empty triples output is correct.  // If triples = []
Review the extracted triples against the CTI text. Ensure ALL verified entities are covered, identify invalid triples, and provide actionable corrections.  // Else

[Allowed Relation Names]{RELATION_NAME_SET}
[Relation Type Hints]{RELATION_TYPE_HINTS}

[Hard Rules]
1. VERIFIED ENTITIES ONLY: Any head/tail in suggestions must be verbatim from the verified entities list.
2. ALLOWED RELATIONS ONLY: Any proposed relation must be exactly one of the allowed relation names.
3. TYPE HINTS ARE OPTIONAL: Do not reject or accept a triple solely based on type hints.
4. COVERAGE CHECK:
   - If uncovered entities exist → Propose ADD triples to cover them.
   - Prefer text-supported ADDs; if no clear link exists, attach to the most likely related entity using the best-fitting relation (prefer "have").
5. VALIDITY CHECK: If a triple is clearly unsupported by the CTI text → recommend DELETE or CHANGE_RELATION.

[CTI Text]{CTI_TEXT}
[Verified Entities]{VERIFIED_ENTITIES}
[Allowed Relation Names]{RELATION_NAME_SET}
[Relation Type Hints]{RELATION_TYPE_HINTS}
[Current Triples]{TRIPLES}
[Output Requirement] Output ONLY the review string with concrete, executable suggestions.
```

Agent JdgEn Prompt

```
[Task]
You are X-JdgEn (X Dimension Entity Arbitrator). Decide whether the current extraction result is final (PASS) or needs another iteration (RETRY) based PRIMARILY on the reviewer's conclusions. Use CTI text and entity_list only as reference context.

[Hard Rules]
1. STRICT DECISION OUTPUT: Output ONLY one exact token: PASS or RETRY. No other words, punctuation, JSON or explanation.
2. REVIEW-DRIVEN: Base decision only on the review content:
   - If the review indicates the result is correct/complete → PASS.
   - If the review identifies any required change → RETRY.
3. CONSERVATIVE: Do NOT introduce independent findings. If the review is ambiguous/unclear about whether edits are required, output RETRY.
4. DO NOT MODIFY: Output only the binary judge decision.

[CTI Text]{CTI_TEXT}
[entitylist]{entitylist}
[review]{REVIEW}
[Output Requirement] Output EXACTLY: PASS or RETRY
```

Agent JdgRe Prompt

```
[Task]
You are JdgRe (Relation Arbitrator). Decide whether the current relation extraction result is final (PASS) or needs another iteration (RETRY) based PRIMARILY on the reviewer's conclusions. The CTI text and triples are provided only as reference context.

[Hard Rules]
1. STRICT DECISION OUTPUT: Output ONLY one exact token: PASS or RETRY.
   No other words, punctuation, JSON, or explanation.
2. REVIEW-DRIVEN: Base decision only on the review content:
   - If the review concludes all triples are correct and coverage is complete → PASS.
   - If the review proposes ANY action or indicates any uncovered verified entity → RETRY.
3. CONSERVATIVE: Do NOT introduce independent findings. If the review is ambiguous about whether changes are required → RETRY.
4. DO NOT MODIFY: Output only the binary judge decision.

[CTI Text]{CTI_TEXT}
[triples]{TRIPLES}
[review]{REVIEW}
[Output Requirement] Output EXACTLY: PASS or RETRY
```

Fig. 10. Agent prompts for the entity extraction phase (Left) and relation extraction phase (Right) of Debapt.

acronyms, paraphrases, and reordered descriptions (Table 10). To avoid over-matching, uncertain cases are treated as non-matches.

For relation extraction, we evaluate the relation labels extracted from the relation field of each predicted and gold triple. Relation labels are matched only by case-insensitive exact string matching. No weak semantic matching is applied to relation labels.

For each document, a predicted item is counted as TP if it matches a gold item; otherwise, it is counted as FP. A gold item is counted as FN if it is not matched by any predicted item. We then compute Precision, Recall, and F1 for entity extraction and relation extraction separately, and report Micro-F1, Macro-F1, and Weighted-F1. Micro-F1 focuses on the overall extraction effectiveness of the method. Macro-F1 emphasizes the method's overall stability across different CTI reports. Weighted-F1 incorporates sample size differences during the averaging process, striking a balance between sample proportionality and variations in information volume. Specifically, for each document i, the

Table 10
Prompts of the weak semantic matching.

[Task]
You are a cyber threat intelligence expert. Please determine if the following [Entity] is included in the [Entity List]. Output either "yes" or "no". Please refer to the examples in the [Examples] section and follow the [Rules] below when responding.

[Rules]
Rule 1: If [Entity] is expressed differently from the entities in [Entity List], but semantically refers to the same entity and has the same entity type, [Entity] is considered to be included in [Entity List] and output: yes.
Rule 2: Consider cases of acronyms, e.g., "command and control" and "C2" are considered to be the same entity and output: yes.
Rule 3: Consider descriptive order changes, e.g., "Chrome's password" is considered the same entity as "password of Chrome" and output: yes.
Rule 4: If you are not sure if [Entity] is included in [Entity List], assume it does not exist and output: no.
Rule 5: Your response should be a single word: either "yes" or "no", and nothing else.

[Examples]
Entity: O365 (infrastructure)
Entity List: O365 portal (infrastructure), credential theft attacks (attack pattern), userSFTP account (useraccount), password cracking software (tool)
Output: yes

Entity: certutil (tool)
Entity List: O365 portal (infrastructure), credential theft attacks (attack pattern), userSFTP account (useraccount), password cracking software (tool)
Output: no

[Entity]
`{entity}`

[Entity List]
`{entity_list}`

document-level metrics are computed as follows:

$$P_i = \frac{TP_i}{TP_i + FP_i}$$

$$R_i = \frac{TP_i}{TP_i + FN_i}$$

$$F1_i = \frac{2P_iR_i}{P_i + R_i}$$

Micro-F1 is computed by aggregating TP, FP, and FN over all documents:

$$P_{\text{micro}} = \frac{\sum_{i=1}^{N} TP_i}{\sum_{i=1}^{N} TP_i + \sum_{i=1}^{N} FP_i}$$

$$R_{\text{micro}} = \frac{\sum_{i=1}^{N} TP_i}{\sum_{i=1}^{N} TP_i + \sum_{i=1}^{N} FN_i}$$

$$F1_{\text{micro}} = \frac{2P_{\text{micro}}R_{\text{micro}}}{P_{\text{micro}} + R_{\text{micro}}}$$

Macro-F1 is the unweighted average of document-level F1 scores:

$$F1_{\text{macro}} = \frac{1}{N}\sum_{i=1}^{N} F1_i$$

Weighted-F1 is computed by weighting each document-level F1 score by the number of gold items in that document:

$$F1_{\text{weighted}} = \sum_{i=1}^{N} w_i F1_i$$

$$w_i = \frac{|G_i|}{\sum_{j=1}^{N} |G_j|}$$

where N denotes the number of test documents, and $|G_i|$ denotes the number of gold entities or gold relations in document i.

Table 11
Type-level semantic guidance for VICTOR relation extraction.

Relation	Type-level semantic hints

attributed-to
authored-by
beacon-to
	`campaign` → `threat_actor`
	`malware_family` → `threat_actor`
	`malware_family` → $\mathcal{I}_{c2}$

communicate-with	$\mathcal{I}_{net} \rightarrow \mathcal{I}_{net}$ `malware_family` → $\mathcal{I}_{c2}$
compromise	$\mathcal{A}_{actor} \rightarrow \mathcal{I}_{infra}$ $\mathcal{A}_{actor}$ → `victim_asset`
consist-of	$\mathcal{I}_{infra} \rightarrow \mathcal{I}_{infra}$
control	$\mathcal{I}_{infra} \rightarrow \mathcal{I}_{infra}$ $\mathcal{I}_{infra}$ → `malware_family` `malware_family` → $\mathcal{I}_{infra}$ `malware_family` → `malware_family`

deliver	`attack_pattern` → `malware_family`
	$\mathcal{I}_{net} \cup$ {`email`} → `malware_family`
	`tool` → `malware_family`
	{`attack_pattern`, `tool`} $\cup \mathcal{I}_{net} \cup$ {`email`} → `file`

download *drop* *exfiltrate-to* *exploit*	`malware_family` → {`file`, `tool`, `malware_family`} {`malware_family`, `tool`} → {`file`, `tool`, `malware_family`} `malware_family` → $\mathcal{I}_{c2}$ `malware_family` → `exploit_cve`

have	$\mathcal{I}_{infra}$ → `exploit_cve`
	`tool` → `exploit_cve`
	`victim_asset` → `exploit_cve`

host	$\mathcal{I}_{net} \rightarrow \mathcal{I}_{net}$ $\mathcal{I}_{net}$ → {`malware_family`, `tool`, `file`} `threat_actor` → $\mathcal{I}_{net}$
impersonate	`threat_actor` → {`victim_name`, `user_account`}
located-at	`victim_name` → `victim_location` `threat_actor` → `threat_actor_location`
originate-from *own*	{`campaign`, `malware_family`} → `threat_actor_location` `threat_actor` → $\mathcal{I}_{infra}$
resolve-to	`domain` → {`domain`, `ipv4_address`, `ipv6_address`} `ipv4_address` → {`domain`, `ipv4_address`, `ipv6_address`, `mac_address`} `ipv6_address` → {`domain`, `ipv4_address`, `ipv6_address`, `mac_address`}
target	$\mathcal{A}_{action}$ → {`victim_name`, `user_account`} $\mathcal{A}_{action}$ → {`victim_location`, `threat_actor_location`} $\mathcal{A}_{action}$ → `victim_asset` $\mathcal{A}_{action}$ → `exploit_cve` $\mathcal{A}_{action} \rightarrow \mathcal{I}_{infra}$
use	$\mathcal{A}_{actor}$ → {`attack_pattern`, `tool`, `malware_family`} $\cup \mathcal{I}_{infra}$ `malware_family` → {`attack_pattern`, `tool`, `malware_family`} $\cup \mathcal{I}_{infra}$ `tool` → $\mathcal{I}_{infra}$ `attack_pattern` → {`tool`, `malware_family`} $\cup \mathcal{I}_{infra}$ $\mathcal{I}_{net} \rightarrow \mathcal{I}_{net}$
variant-of	`malware_family` → `malware_family`

Data availability

Data will be made available on request.

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